The Sensorium Manifold:

Emergent Modes of Intelligence

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Abstract

Current generative architectures treat sequence modeling as an initial value problem: gradient-trained weights predict the next token before any future constraint can be applied. We reject this framing entirely and cast generation as a *boundary value problem* (BVP). The observer clamps a subset of symbols as fixed constraints and the remaining degrees of freedom are resolved simultaneously via massively parallel geometric resonance search.

We instantiate this paradigm in a physics-native substrate built on three layers:

1. An **8191-bit Value** representation over the Mersenne field GF(8191) ($2^{13} - 1$), where each byte value maps to a deterministic prime-basis bitset across 128 blocks of 64 bits, and bitwise operations (`OR`, `AND`, `popcount`) execute discrete wave interference at single-cycle throughput.
2. A **Morton-coded Radix Trie** (Distributed Merkle Trie) that indexes values by their spatial keys, with sequential structure encoded as composable affine transformations over GF(8191).
3. An **Analytical Phase** navigation layer that maps values onto a $S^1 \times S^1$ phase torus via their intrinsic prime-index geometry over GF(8191), enabling deterministic closure detection and intent steering without trained co-occurrence models.

No loss function is minimized. No weights are updated. Inference is a single-pass GF(8191) geometric distance scan across the entire memory substrate, resolving the nearest rotational state via integer arithmetic on GPU. A two-phase hybrid retrieval pipeline (bitwise filter followed by complex-valued PhaseDial scoring) provides precise candidate ranking. A self-programming loop closes the system: the cantilever BVP solver synthesizes missing operators, successful bridges harden into native macro-opcodes, and the interpreter dispatches them directly—reifying execution traces into new operators. We validate the architecture across text classification, code generation, logical reasoning, phase-dial generalization, image reconstruction, and cross-modal tasks, reporting both successes and failure regimes with equal detail.

0.1 Motivation

The current generative AI stack is structurally constrained by its formulation as an *initial value problem*. Whether through autoregressive transformers or state-space models, generation is treated as strictly sequential extrapolation: given a prefix $x_{0:t}$, compute x_{t+1}, append it, and repeat. This framing imposes three fundamental limitations:

1. **Serial depth bottleneck.** Each step depends on the materialization of all preceding outputs, forcing an $O(N)$ critical-path depth that hardware parallelism cannot bypass.
2. **Irrevocable commitment.** Tokens cannot be retracted once emitted; local errors propagate forward with no mechanism for future constraints to resolve past ambiguity.
3. **Static memory.** Knowledge is compressed into static weight matrices requiring gradient descent to update, causing catastrophic forgetting and forcing reliance on retrieval augmentation.

This paper proposes a departure from the IVP framework. We cast generation as a *boundary value problem*: the observer clamps constraints, and the remaining degrees of freedom are resolved simultaneously by a massively parallel geometric resonance engine operating over the Mersenne field GF(8191).

0.2 Why Bitwise Math Is Wave Interference

The key insight underlying this architecture is that binarized prime-frequency vectors are not merely compressed representations—they are discrete Fourier transforms of semantic waves. Each bit in an 8191-bit Value represents the presence of a specific fundamental oscillator, packed into 128 blocks of 64 bits. Bitwise operations then execute actual discrete wave interference:

- `OR` **(Superposition):** Combining tokens in a context window $(A \mid B)$ superposes their oscillator spectra.
- `popcount`$(A \,\&\, B)$ **(Constructive Interference):** Measures the amplitude of shared oscillators between context and memory.
- `popcount`$(A \,\&\, {\sim}B)$ **(Destructive Interference / Noise):** Measures oscillators present in memory but absent from context—harmonic noise or phase mismatch.

Mathematically, the dot product of two binarized vectors equals the popcount of their bitwise AND. Each comparison requires $\lceil 8191/64 \rceil = 128$ integer operations—the architecture compiles wave mechanics down to 128 single-cycle machine-word operations per value pair.

0.3 Paper Organization

The remainder of this paper is organized as follows. Section 0.13 defines the 8191-bit Value primitive and its algebraic operations over GF(8191). Section 0.24 introduces the manifold layout, Morton-coded spatial indexing, and affine transformations. Section 0.31 describes the Universal Tokenizer and topological boundary detection. Section 0.18 presents the analytical toroidal phase navigation model. Section 0.4 details the GF(8191) resolver pipeline. Section 0.10 formalizes generation as a boundary value problem and the cantilever solver. **??** describes prime-field storage in the Morton-coded radix trie. Section 0.34 presents the virtual machine, macro-opcode hardening, and the self-programming loop. Section 2 describes topological entropy routing and virtual mitosis. Section 1.1 introduces the dual Substrate/Graph architecture. Experimental results span text classification, code generation, logical reasoning (bAbI), phase-dial robustness, image reconstruction, cross-domain completion, scaling analysis, and Graph channel routing.

0.4 Problem Statement

Given a context GF(8191) rotation $\mathbf{q} = (a_q, b_q)$ and a graph of N nodes each carrying their own rotation state $\{(a_i, b_i)\}_{i=1}^{N}$, the resolver returns:

$$i^* = \underset{1 \le i \le N}{\arg\min}\; d^2(\mathbf{q}, \mathbf{n}_i) \tag{1}$$

where d^2 is the squared Euclidean distance in the Galois field coordinate space. This is the system's sole inference primitive—all generation, retrieval, and reasoning reduce to repeated resolver calls.

0.5 GF(8191) Geometric Distance

Each node in the thermodynamic field carries a composable affine state $(a, b) \in \mathrm{GF}(8191)^2$ where:

$$f(x) = (a \cdot x + b) \bmod 8191, \qquad a \in \{1, \ldots, 8190\},\ b \in \{0, \ldots, 8190\} \tag{2}$$

The resolver computes the distance between a query context and every candidate node as:

$$d^2(\mathbf{q}, \mathbf{n}_i) = (a_i - a_q)^2 + (b_i - b_q)^2 \tag{3}$$

This is pure integer arithmetic—no floating-point operations, no popcount, no matrix multiplies. The geometric interpretation is direct: a encodes the *multiplicative* (scaling/rotation) identity, b encodes the *additive* (translational) offset. Nearby nodes in GF(8191) space share similar rotational histories.

0.6 Single-Pass Parallel Scan

The resolver executes as a single-pass parallel scan across all N nodes:

1. Each GPU thread evaluates $d^2(\mathbf{q}, \mathbf{n}_i)$ for its assigned node (2 subtractions, 2 multiplications, 1 addition).
2. The distance is inverted to enable lock-free maximum selection: $\hat{s}_i = 131{,}072 - d_i^2$, where 131,072 is the packed distance ceiling configured in the compute kernel.
3. The score and index are packed into a single `uint64`:

$$\text{packed}_i = \big(\hat{s}_i \ll 32\big) \bigm| i \tag{4}$$

The winner is determined by a single `AtomicMax` operation per thread. Because the inverted distance occupies the high 32 bits, lexicographic integer comparison correctly preserves distance ordering.

Computational Cost. Each thread executes exactly 5 integer operations (2 subtracts, 2 multiplies, 1 add) plus one atomic compare-and-swap. The entire scan is memory-bound at 4 bytes per candidate (one `GFRotation` = 2 × `uint16`).

0.7 Multi-Backend Dispatch

The kernel dispatch layer supports Metal (Apple Silicon), CUDA (NVIDIA), and CPU (fallback) backends with automatic selection via the `SIX_BACKEND` environment variable:

$$i^* = \underset{i \in \text{Nodes}}{\arg\min}\ d^2(\mathbf{q}, \mathbf{n}_i) \tag{5}$$

For CUDA, multi-GPU orchestration distributes the node array across devices with per-device base offsets, merging local winners via the packed score representation.

0.8 Composition Instead of Accumulation

A key distinction from popcount-based retrieval is that the GF(8191) resolver operates on *composed* rotational states rather than accumulated bit densities. When the Sequencer detects a topological boundary event, it emits a rotation:

- **Density Spike** → X_{90}: $(1, 1)$
- **Phase Inversion** → Y_{90}: $(g, 0)$
- **Density Trough** → Z_{90}: $(g, 1)$
- **Low-Variance Flux** → X_{180}: $(1, 2)$

where g is the smallest primitive root of GF(8191), computed at runtime. These compose via the group operation:

$$(a_2, b_2) \circ (a_1, b_1) = \big((a_2 \cdot a_1) \bmod 8191,\ \ (a_2 \cdot b_1 + b_2) \bmod 8191\big) \tag{6}$$

Because composition is non-commutative, the resolver distinguishes sequences that traverse different rotational paths even if they contain identical content—encoding syntax as geometry.

0.9 The Autoregressive IVP Limitation

Contemporary language models formulate generation as an autoregressive Initial Value Problem (IVP): given a starting context, continuously predict the next token. Formally, given prefix $\mathbf{x}_{0:t}$, the IVP computes $p(x_{t+1} \mid \mathbf{x}_{0:t})$ and samples, yielding a Markov chain with strictly left-to-right information flow. This creates three structural limitations:

1. **Serial depth bottleneck:** Each step depends on the materialization of all preceding outputs, forcing an $O(N)$ critical-path depth that hardware parallelism cannot bypass.
2. **Irrevocable commitment:** Tokens cannot be retracted once emitted; local errors propagate as distributional drift—a single decoding error permanently alters the generation trajectory.
3. **Static memory:** Knowledge is compressed into continuous weight matrices $\mathbf{W}$ requiring gradient descent to update, causing catastrophic forgetting.

0.10 The Cantilever BVP Formulation

We reject the IVP formulation entirely, framing generation instead as a *Boundary Value Problem (BVP) solved through coordinate geometry.* Given a boundary constraint (a starting anchor coordinate) and a distant logic goal (a target phase ϕ_t), the generation engine extends a computational cantilever into unknown territory to bridge the gap.

Generation relies purely on interpolation of rotational trajectories through GF(8191) rather than predicting the next localized step. The Interpreter carries execution registers to synthesize temporary affine momentum and phase states without committing to the radix trie, mathematically ensuring that paths can diverge, extrapolate, and backtrack gracefully.

0.11 The Frustration Engine

When the cantilever wavefront stalls—meaning no locally available sequence matches the momentum dictated by the execution register—it enters a state of *frustration*. Rather than halting or hallucinating, the system evaluates the unresolved rotation gap between the target phase ϕ_t and the current locked phase ϕ_c:

$$\Delta = \phi_t \cdot \phi_c^{-1} \pmod{8191} \tag{7}$$

When $\Delta = 1$, the span has successfully bridged to the target (phase-lock). When $\Delta \neq 1$, the wavefront elevates frustration levels and searches the *MacroIndex*—a registry of discovered affine operators indexed by their full geometric AffineKey (one 64-bit word per Value block). The MacroIndex supports two resolution modes:

1. **Exact lookup:** The geometric delta between start and goal boundaries is computed as an AffineKey and matched directly against the stored opcode table. A hit returns the corresponding MacroOpcode with its affine operator $f(x) = \textit{scale} \cdot x + \textit{translate} \pmod{8191}$.
2. **Approximate lookup:** When no exact match exists, the AffineKey is projected into PhaseDial space—a 512-dimensional unit-normalized complex vector derived from the key's structural phases. Cosine similarity is computed against PhaseDial embeddings of all hardened opcodes. The nearest neighbor above a configurable threshold (default 0.7) is returned as an approximate bridge operator.

If a MacroOpcode resolves the missing affine rotation, the wavefront seamlessly applies the learned operator to satisfy the boundary constraint.

0.12 The Cantilever Generation Loop

Generation extends the originating seed like a physical cantilever through a multi-stage pipeline:

1. **Tokenize:** Prompt bytes are fed to the Universal Tokenizer via `WriteBatch` RPC, producing Morton-packed keys that are compiled into a `primitive.Value` sequence through the same path used for corpus ingestion.
2. **Stage 1 — Exact lexical continuation:** The prompt Value sequence is decoded to bytes and matched against the stored corpus. For each corpus row, the system checks whether the decoded prompt is an exact prefix of the row's byte sequence; the first match yields the suffix as a continuation.
3. **Stage 2 — Operator-mediated bridging:** When no exact prefix match exists, the system performs BVP bridging. For each corpus row longer than the prompt:
   (a) The prompt's OR-folded signal serves as the start boundary ϕ_s.
   (b) The row's suffix signal serves as the goal boundary ϕ_g.
   (c) `BridgeValues` computes the AffineKey delta and queries the MacroIndex (exact, then approximate) for an operator that resolves the geometric gap.
   (d) If a bridge operator is found, it is applied to the start phase. The pre-residue (start active count) and post-residue (distance between bridged phase and goal) are recorded.
4. **Harden:** Every successful Stage 2 bridge feeds `RecordCandidateResult` with the bridge outcome (pre-residue, post-residue, advanced flag, stable flag). Candidates that accumulate ≥ 5 consecutive successes where post-residue $<$ pre-residue are promoted to Hardened status, embedding their AffineKey into PhaseDial space for future approximate retrieval.
5. **Emit:** The continuation—either the exact suffix or the bridged suffix—is decoded from Values back to bytes via `InferLexicalSeed` and returned as the generation output. Geometric closure occurs when the resulting coordinate state achieves phase-lock ($\Delta = 1$) against the target boundary.

0.13 Deterministic Prime-Basis Identity

Let $D = 8191$ denote the value dimensionality (the Mersenne prime $2^{13} - 1$) and let $\mathcal{B} = \{0, \ldots, 255\}$ be the byte alphabet. Every byte value $b \in \mathcal{B}$ is mapped to a deterministic binary vector $\mathbf{c}(b) \in \{0,1\}^D$ by activating exactly five bits at positions determined by a Golomb-ruler basis projected through coprime multipliers modulo the Mersenne prime $D = 8191$.

Let $\mathcal{G} = \{0, 1, 4, 9, 11\}$ be the 5-sparse Golomb-ruler basis and let $\{m_1, \ldots, m_5\}$ be dynamically derived coprime multipliers (`byteValueMultipliers`). The activation set is:

$$\mathcal{P}(b) = \left\{ g_j \cdot m_j \cdot b \bmod D \;\middle|\; g_j \in \mathcal{G},\ j = 1, \ldots, 5 \right\} \tag{8}$$

$$c_i(b) = \begin{cases} 1 & \text{if } i \in \mathcal{P}(b), \\ 0 & \text{otherwise.} \end{cases} \tag{9}$$

Because $D = 8191$ is a Mersenne prime ($2^{13} - 1$), every non-zero multiplier generates a full-period permutation of the residues $\{0, \ldots, D-1\}$, and the Golomb-ruler property guarantees that no two basis offsets share the same pairwise difference. Together these ensure that distinct byte values receive distinct activation patterns (verified by exhaustive enumeration over all 256 values). The resulting value is extremely sparse: $\|\mathbf{c}(b)\|_1 = 5$ out of $D = 8191$ bits, yielding a sparsity ratio of $5/8191 \approx 0.061\%$.

Why Coprime Multipliers over a Golomb Ruler. The choice of coprime multipliers is not arbitrary. Multiplying by a value p and reducing modulo a prime $D = 8191$ generates a permutation of the residues $\{0, \ldots, D-1\}$ for any $p \not\equiv 0 \pmod{D}$. Each dynamically derived multiplier independently generates a distinct full-period permutation. The Golomb-ruler basis $\{0, 1, 4, 9, 11\}$ further guarantees that all pairwise differences between basis positions are unique, maximizing the minimum Hamming distance between any two byte values across the full 8191-bit space.

0.14 Algebraic Operations

The Value supports a closed algebra of compositional operations. Each executes in $O(D/w)$ time where w is the SIMD word width (64 for `uint64`, yielding $\lceil D/w \rceil = 128$ operations per value):

$$\textbf{ValueOR:} \quad (\mathbf{a} \vee \mathbf{b})_i = a_i \vee b_i \qquad \text{(Superposition / LCM)} \tag{10}$$

$$\textbf{ValueGCD:} \quad (\mathbf{a} \wedge \mathbf{b})_i = a_i \wedge b_i \qquad \text{(Shared factors / Intersection)} \tag{11}$$

$$\textbf{ValueHole:} \quad H(\mathbf{t}, \mathbf{e})_i = t_i \wedge \neg e_i \qquad \text{(Structural Vacuum)} \tag{12}$$

$$\textbf{Similarity:} \quad \mathrm{sim}(\mathbf{a}, \mathbf{b}) = \sum_{i=0}^{D-1} a_i \cdot b_i = \texttt{popcount}(\mathbf{a} \wedge \mathbf{b}) \qquad \text{(Discrete dot product)} \tag{13}$$

Equivalence to Inner Product. For binary vectors $\mathbf{a}, \mathbf{b} \in \{0,1\}^D$, the real-valued inner product $\langle \mathbf{a}, \mathbf{b} \rangle$ reduces exactly to the popcount of their bitwise AND:

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=0}^{D-1} a_i \, b_i = \big|\{i : a_i = 1 \wedge b_i = 1\}\big| = \texttt{popcount}(\mathbf{a} \,\&\, \mathbf{b}) \tag{14}$$

This identity is the mechanism by which single-cycle integer instructions execute semantic similarity computation.

0.15 Positional Binding via Circular Shift

To encode sequential position without destroying content, we define the *RollLeft* operator as a circular permutation over bit indices:

$$\mathrm{RollLeft}(\mathbf{c}, \, s)_i = c_{(i-s) \bmod D} \tag{15}$$

Because circular shifts are invertible and non-commutative with superposition, positional binding preserves all algebraic properties while ensuring that re-ordered sequences produce distinct values:

$$\mathrm{RollLeft}(\mathbf{c}_d, 0) \vee \mathrm{RollLeft}(\mathbf{c}_o, 1) \vee \mathrm{RollLeft}(\mathbf{c}_g, 2) \;\neq\; \mathrm{RollLeft}(\mathbf{c}_g, 0) \vee \mathrm{RollLeft}(\mathbf{c}_o, 1) \vee \mathrm{RollLeft}(\mathbf{c}_d, 2) \tag{16}$$

That is, “`dog`” $\neq$ “`god`” in the value representation.

0.16 SimHash Structural Binning

The *ValueBin* operator maps any value $\mathbf{c}$ to an 8-bit structural bin $b \in \{0, \ldots, 255\}$ via a SimHash-style random hyperplane projection. Given $k = 8$ fixed seed values $\{s_1, \ldots, s_8\}$, the accumulator for plane j sums signed contributions from each active bit:

$$\mathrm{acc}_j = \sum_{i : c_i = 1} \begin{cases} +1 & \text{if } h(i, s_j) \geq 2^{63}, \\ -1 & \text{otherwise,} \end{cases} \qquad h(i, s_j) = i \cdot s_j + (i \ll (j+1)) \tag{17}$$

The bin is then the 8-bit integer whose j-th bit is $[\mathrm{acc}_j \geq 0]$. Values with high Hamming similarity map to the same or nearby bins with high probability, enabling the EigenMode phase table (Section 0.18) to use a fixed-size 256×256 co-occurrence matrix without per-value storage.

0.17 FlatValue: GPU-Optimized Dense Representation

For GPU dispatch, the sparse 8191-bit Value is converted to a *FlatValue*: a dense array of at most D active prime indices $\{p_1, \ldots, p_k\}$ where $k = \|\mathbf{c}\|_1$. This eliminates bit-twiddling thread divergence in SIMT architectures: every thread iterates over exactly k entries rather than scanning all $\lceil D/64 \rceil = 128$ blocks.

Batched flattening across N values is parallelized via a pre-warmed, auto-scaling worker pool, yielding amortized $O(k)$ per value with near-zero goroutine creation overhead.

0.18 Analytical Phase Model

Unlike spectral methods that require building co-occurrence matrices and extracting eigenvectors, the phase model derives toroidal coordinates *analytically* from the intrinsic bit distribution of each value over GF(8191). No training data is required; every value's position on the phase torus is determined at construction time.

0.18.1 Phase Assignment via Prime Geometry

Each value $\mathbf{c}$ has $k = \|\mathbf{c}\|_1$ active bits at positions $\{p_1, \ldots, p_k\} \subset \{0, \ldots, 8190\}$. We assign toroidal coordinates $(\theta, \phi) \in [-\pi, \pi)^2$ directly:

Theta (Circular Mean of Active Positions). The angular identity of a value is the circular mean of its active prime indices, uniformly distributed around $[0, 2\pi)$ over the 8191-element field:

$$\theta(\mathbf{c}) = \operatorname{atan2}\Big(\sum_{j=1}^{k} \sin\big(\tfrac{2\pi\, p_j}{8191}\big),\ \sum_{j=1}^{k} \cos\big(\tfrac{2\pi\, p_j}{8191}\big)\Big) \tag{18}$$

Values with similar prime activation patterns occupy nearby θ-positions on the torus, making the circular mean a natural geometric coordinate for the GF(8191) value space.

Phi (Density Phase). The secondary phase encodes the value's population density:

$$\phi(\mathbf{c}) = \frac{2\pi \cdot \|\mathbf{c}\|_1}{8191} \tag{19}$$

Sparse values (low popcount) sit near $\phi = 0$; dense values approach $\phi = 2\pi$. The pair (θ, ϕ) embeds every value onto the 2-torus $\mathbb{T}^2 = S^1 \times S^1$.

0.19 Sequential Phase Tracking and Intent Alignment

During generation, the accumulated sequential phase is tracked as a *weighted circular mean* over the emitted value sequence $(\mathbf{c}_1, \ldots, \mathbf{c}_n)$:

$$\bar{\theta}_n = \operatorname{atan2}\Big(\sum_{i=1}^{n} w_i \sin\theta(\mathbf{c}_i),\ \sum_{i=1}^{n} w_i \cos\theta(\mathbf{c}_i)\Big) \tag{20}$$

where $w_i = \|\mathbf{c}_i\|_1$ weights each value by its active count (density). The *concentration parameter* κ_n (analogous to the κ parameter of the von Mises distribution) measures phase coherence:

$$\kappa_n = \frac{\sqrt{S_n^2 + C_n^2}}{W_n}, \qquad S_n = \sum_{i=1}^{n} w_i \sin\theta_i, \quad C_n = \sum_{i=1}^{n} w_i \cos\theta_i, \quad W_n = \sum_{i=1}^{n} w_i \tag{21}$$

Values of κ_n near 1 indicate strong phase alignment (the trajectory is coherent); values near 0 indicate phase diffusion (semantic drift).

0.20 Geometric Closure Detection

Generation terminates when the candidate value sequence's weighted circular mean phase returns to within a configurable threshold δ of the anchor phase $\bar{\theta}_{\text{anchor}}$ established at prompt initialization:

$$\Delta\theta = \min\bigl(|\bar{\theta}_n - \bar{\theta}_{\text{anchor}}|,\ \ 2\pi - |\bar{\theta}_n - \bar{\theta}_{\text{anchor}}|\bigr) < \delta \tag{22}$$

where δ is derived from the architecture's Shannon capacity parameter (`shannonCapacity`, default 0.45, a density threshold). This constitutes a topological proof of structural closure: the generative trajectory has completed a semantically coherent loop on the torus. Because $\mathbb{T}^2$ is a compact manifold, every trajectory is pre-compact (bounded), preventing entrapment in isolated fixed points or periodic orbits.

0.21 Advantages of the Analytical Model

- **Zero training cost.** No co-occurrence matrices, no eigendecomposition, no corpus-dependent calibration. The phase map is a pure function of the value's bit pattern.
- **Deterministic.** Identical values always map to identical (θ, ϕ) regardless of context, ensuring reproducibility.
- $O(k)$ **per value.** Phase computation scales with the number of active bits ($k \leq 8191$), not with corpus size.
- **Intrinsically geometric.** The prime-index circular mean directly corresponds to the value's position in the GF(8191) rotational lattice, making phase navigation and Galois field operations composable.

0.22 Two-Phase Retrieval Pipeline

The system resolves queries through a unified pipeline that hybridizes bitwise structural measurement with scalar algebraic cancellation over GF(8191):

1. **Phase 1 — Resonance Measurement (XOR):** Incoming queries project a rotational trajectory into the Address Plane. The projected state is evaluated against stored anchors using bitwise XOR as a pure scalar measurement of Hamming-space distance:

$$\text{distance}(A, B) = \texttt{popcount}(A \oplus B) \tag{23}$$

   This measurement filters for paths experiencing constructive interference (low residue). Crucially, XOR is strictly an observable measurement gauge; it is never used to permanently compose or store state, as that would inject uncontrolled Shannon entropy.
2. **Phase 2 — Algebraic Cancellation:** Once a resonant geometric cluster is localized, logical reasoning over the localized multi-context merges resolves via scalar modular arithmetic. Because multiple facts coexist in a single cell via additive superposition $\phi_{\text{merged}} = (\phi_A + \phi_B) \pmod{8191}$, querying isolated entities involves inverse cancellation:

$$\text{result} = \phi_{\text{merged}} \cdot \phi_A^{-1} \pmod{8191} \approx \phi_B \tag{24}$$

3. **Phase 3 — Frustration or Readout:** If result matches a valid anchor phase ($\Delta = 1$), the wavefront locks and proceeds. Otherwise, the gap triggers *frustration*, signaling the need for an extrapolated tool or a backtrack pivot.

0.23 Destructive Interference (Negative Constraints)

Beyond positive phase-locking, the dual algebra naturally enforces negative constraints without heuristic branching rules. A negated fact (e.g., "Roy is NOT in the kitchen") is encoded as the absolute additive inverse in the field:

$$\phi_{\text{neg}} = (8191 - \phi_{\text{pos}}) \pmod{8191} \tag{25}$$

When the exploring wavefront encounters ϕ_{neg}, the positive baseline momentum meets the stored anti-phase. The sum algebraically collapses to zero ($\phi_{\text{pos}} + \phi_{\text{neg}} \equiv 0 \pmod{8191}$), immediately eliminating path energy and pruning the search space via pure destructive interference.

0.24 Structural Layout: The Address Plane

The architecture utilizes a unified address plane in the form of a Morton-keyed radix trie (DMT). A cell address packs the byte identity (X) and the boundary-local depth (Y) into a single 64-bit key:

$$\text{CellKey} = \text{Pack}(\underbrace{v}_{\text{byte value (X)}}, \underbrace{d}_{\text{local depth (Y)}}) \tag{26}$$

The primary layout is computationally efficient (`symbol` $\ll 32 \mid$ `localDepth`), ensuring that data scan locality intrinsically follows the underlying sequential stream. Because the depth d resets to 0 at each structural boundary discovered by the sequencer, distinct sequences that share identically situated local depths collapse geometrically onto the exact same cell coordinate.

0.25 The Native Monotype

At each defined Morton coordinate, the system instantiates its fundamental computational primitive: an 8384-bit *Value* spanning 131 64-bit blocks. Unlike traditional symbol embeddings, the Value does not redundantly encode the byte (which is already encoded by the spatial X coordinate), but rather serves as a self-contained operational state.

The blocks are structurally partitioned to separate the generative logic from routing metadata:

- **Core (128 blocks, 8192 bits):** The GF(8191) Mersenne core. Each of the 8191 addressable bit positions corresponds to an element of the prime field $\mathbb{F}_{8191}$ where $8191 = 2^{13} - 1$. This bit-field acts as the mathematical reasoning state, participating in core affine evolutions. The final core block is masked to the field boundary.
- **Shell (3 blocks, 192 bits):** Three 64-bit words that operate beyond the GF(8191) plane:
  - *Block 0 (residual/carry):* Stores the logical GF(8191) state phase as a scalar for fast retrieval without scanning the core.
  - *Block 1 (opcode):* Encodes the current operation, delimiter behaviour, and control flow.
  - *Block 2 (operator):* Packs the affine operator (scale and translate, 13 bits each), trajectory endpoints (from/to, 13 bits each), guard radius (7 bits), active flag (1 bit), and status flags (4 bits) into a single 64-bit word.

This layout occupies $131 \times 64 = 8384$ bits of storage per Value, with the 128-block core aligned to cache-line boundaries for zero-cost metadata propagation alongside the core mathematical updates.

0.26 Generative Coordinate Transforms

With the static sequence completely decoupled into the radix trie coordinates, reasoning progresses strictly through generative transform logic within the Mersenne core array. The core performs affine phase rotation over the cyclic GF(8191) group:

$$f(x) = (a \cdot x + b) \pmod{8191}, \qquad a \in \{1, \ldots, 8190\},\ b \in \{0, \ldots, 8190\} \tag{27}$$

Multiplication modulo a Mersenne prime admits an efficient two-step bit reduction: for $M = 2^{13} - 1$, compute $x \bmod M = (x \mathbin{\&} M) + (x \gg 13)$ and iterate until the result fits within 13 bits. This replaces a general-purpose division with shift-and-add, making the affine step essentially free on modern hardware.

By composing sequences of these rotations, the wavefront maps logical deductions strictly into deterministic spatial leaps across the Morton-keyed address plane.

0.27 The Forest: Concurrent Immutable Radix Trees

Storage is realized by the Distributed Merkle Trie (DMT), a Forest of immutable radix trees that stores Value entries keyed by Morton-coded spatial keys. Each tree in the Forest is a persistent, copy-on-write radix trie built on top of immutable data structures, enabling concurrent reads without locking or coordination.

- **Insertion:** Values are merged into the active tree by writing at the Morton key computed from the token's byte value, position, and scale depth. Writes are serialized through the active tree's transaction mechanism; the immutable backing structure ensures that concurrent readers never observe a partial update.
- **Read routing:** The Forest maintains running latency metrics for each tree. A read request is dispatched to all trees in parallel; the first result to return wins. This races trees against each other, routing traffic to the most performant replica without explicit load-balancing heuristics.
- **Synchronization:** After every insertion, the Forest propagates the update to all sibling trees via a background synchronization loop. Because each tree is immutable at the snapshot level, synchronization is a merge of root pointers rather than a diff of mutable state.

0.28 Morton-Coded Spatial Indexing

Each token is assigned a deterministic 64-bit *Morton key* that packs three coordinates into a single integer for locality-preserving sort order:

$$\mathcal{M}(z, \text{sym}, \text{pos}) = \big(\texttt{uint64}(z) \ll 56\big) \,\big|\, \big(\texttt{uint64}(\text{sym}) \ll 32\big) \,\big|\, \texttt{uint64}(\text{pos}) \tag{28}$$

The bit-field layout from MSB to LSB is:

Bits 63–56	**Bits 55–32**	**Bits 31–0**
z (Scale Index) 8 bits	Symbol (Byte Value) 24 bits	Position (Sequence Offset) 32 bits

This layout is load-bearing in three ways:

1. **Scale stratification:** Grouping by z in the MSBs ensures that all entries at the same hierarchical depth sort contiguously, enabling prefix-bounded iteration within a scale layer.
2. **Symbol locality:** Within a scale, all occurrences of the same byte value share a common key prefix, allowing `QueryByByte` to narrow a large store to a single prefix-bounded subtree.
3. **Positional contiguity:** Within a symbol band, positions are numerically ordered in the LSBs, making neighborhood queries (`QueryNeighborhood`) a simple prefix scan.

0.29 Zipfian Distributions and Collision-as-Compression

Natural data sources—text, images, audio—universally exhibit *Zipfian* frequency distributions: a small number of byte-position patterns account for the overwhelming majority of occurrences. The radix trie exploits this directly.

When two tokens share the same Morton key (identical byte value at the same scale and position), the trie insertion natively *coalesces* them into a single entry. This is not a hash collision requiring resolution—it is lossless deduplication:

$$\text{Compression Ratio} = \frac{\text{Raw Token Count}}{\text{Unique Morton Keys}} = \frac{N}{|\{\mathcal{M}(z_i, s_i, p_i)\}_{i=1}^{N}|} \tag{29}$$

Because Zipf's Law guarantees that the frequency of the r-th most common pattern scales as $f(r) \propto r^{-\alpha}$ with $\alpha \approx 1$, the number of unique keys grows sub-linearly with corpus size. Empirically, we observe that compression ratios **increase** as datasets grow larger, because larger corpora sample the tail of the Zipfian distribution more densely, producing proportionally more collisions per novel entry.

Table 1: Collision-compression metrics across datasets. Ratios increase with dataset size, consistent with Zipfian redundancy scaling.

Dataset	**Raw Tokens**	**Unique Keys**	**Dedup. %**	**Compression ×**
CIFAR-10 (60K images)	195,000,000	3,693,182	98.1%	52.8×
MBPP (Python)	—	—	—	—
bAbI (QA)	—	—	—	—
Combined Corpus	—	—	—	—

The theoretical lower bound on compression is the dataset's Shannon entropy H: the system cannot compress below the information content of the source distribution. The gap between observed compression and the entropy bound measures the efficiency of the Morton key design.

0.30 Entropy-Driven Sequence Boundaries

When the Sequencer detects a topological boundary and the active value carries signal (non-empty bit-field), the trie commits the current sequence and advances to a fresh one. The Z-coordinate of the Morton key tracks scale depth: each sequence reset increments z, stratifying memory into episodic layers that naturally encode hierarchical structure.

0.31 The Universal Tokenizer

The Universal Tokenizer converts raw byte streams into geometric tokens without learned vocabularies or modality-specific encoders. Each byte occurrence at position *pos* with scale index z and symbol value s produces one Token:

$$\text{TokenID} = \mathcal{M}(z, s, pos) \tag{30}$$

where $\mathcal{M}$ is the Morton code (Equation (28)). The corresponding Value is the deterministic $\mathbf{c}(s)$ from Equation (8), bound to its sequential position via RollLeft (Equation (15)).

From the moment data is tokenized, the system never examines raw byte values again until final output rendering. All downstream computation operates exclusively on Values and TokenIDs.

0.32 The Topological Sequencer

The Sequencer segments incoming data streams into sequences based on *topological properties* of the data itself, not on corpus-derived statistics (BPE) or fixed window sizes.

Step 1: Measurement. For each token at position t with value $\mathbf{c}_t$, the Sequencer extracts two geometric signals:

$$\text{pop}_t = \|\mathbf{c}_t\|_1 = \texttt{popcount}(\mathbf{c}_t) \qquad \text{(Geometric population count)} \tag{31}$$

$$\text{phase}_t = \sqrt{\Theta_{b_t}^2 + \Phi_{b_t}^2} \qquad \text{(Toroidal phase magnitude)} \tag{32}$$

where $b_t = \text{ValueBin}(\mathbf{c}_t)$ and Θ, Φ are the EigenMode phase tables (Section 0.18).

Step 2: Golden Ratio Exponential Smoothing. Running estimates are maintained via exponential moving averages with the golden ratio as the decay constant:

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad \alpha = \varphi^{-3} \approx 0.236 \tag{33}$$

$$\overline{\mathrm{pop}}_t = (1-\alpha)\,\overline{\mathrm{pop}}_{t-1} + \alpha \cdot \mathrm{pop}_t \tag{34}$$

$$\overline{\mathrm{phase}}_t = (1-\alpha)\,\overline{\mathrm{phase}}_{t-1} + \alpha \cdot \mathrm{phase}_t \tag{35}$$

The golden ratio is not arbitrary: φ^{-3} provides a smoothing constant that balances responsiveness to structural changes against noise suppression, with the irrational decay rate preventing aliasing with any periodic structure in the data.

Step 3: Welford's Online Variance. The absolute deviations $\Delta_{\mathrm{pop}} = |\mathrm{pop}_t - \overline{\mathrm{pop}}_t|$ and $\Delta_{\mathrm{phase}} = |\mathrm{phase}_t - \overline{\mathrm{phase}}_t|$ are tracked using Welford's single-pass online variance algorithm, which maintains running mean $\bar{x}_n$ and sum of squared deviations $M_{2,n}$:

$$\delta_n = \Delta_n - \bar{x}_{n-1} \tag{36}$$

$$\bar{x}_n = \bar{x}_{n-1} + \frac{\delta_n}{n} \tag{37}$$

$$M_{2,n} = M_{2,n-1} + \delta_n \cdot (\Delta_n - \bar{x}_n) \tag{38}$$

The unbiased sample standard deviation is $\hat{\sigma}_n = \sqrt{M_{2,n}/(n-1)}$.

Step 4: Dynamic Z-Score Thresholding. Boundary events are triggered when absolute deviations exceed adaptive thresholds derived from the running variance and shared calibrator sensitivity targets $\gamma_{\mathrm{pop}}, \gamma_{\mathrm{phase}}$:

$$\tau_{\mathrm{pop}} = \bar{\Delta}_{\mathrm{pop}} + \hat{\sigma}_{\mathrm{pop}} \cdot \gamma_{\mathrm{pop}} \tag{39}$$

$$\tau_{\mathrm{phase}} = \bar{\Delta}_{\mathrm{phase}} + \hat{\sigma}_{\mathrm{phase}} \cdot \gamma_{\mathrm{phase}} \tag{40}$$

An additional hard topological boundary fires when:

$$\Delta_{\mathrm{phase}} > \frac{\pi}{\kappa} \tag{41}$$

where κ is the `FrequencySpread` hyperparameter.

Step 5: Event Classification. Boundary events are classified into topological triggers that drive GF(8191) affine rotations of the manifold (Section 0.24):

Event	**Condition**	**Manifold Action**
DensitySpike	$\Delta_{\mathrm{pop}} > \tau_{\mathrm{pop}}$ and $\mathrm{pop}_t > \overline{\mathrm{pop}}_t$	Affine rotation $(1, 1)$
DensityTrough	$\Delta_{\mathrm{pop}} > \tau_{\mathrm{pop}}$ and $\mathrm{pop}_t < \overline{\mathrm{pop}}_t$	Affine rotation $(g, 1)$
PhaseInversion	$\Delta_{\mathrm{phase}} > \tau_{\mathrm{phase}}$ or $\Delta_{\mathrm{phase}} > \pi/\kappa$	Affine rotation $(g, 0)$
LowVarianceFlux	coherence time > 10	Affine rotation $(1, 2)$

0.33 Closed-Loop Calibration

The Sequencer incorporates bidirectional feedback at two timescales:

- **High-frequency (local):** When a boundary fires, the local value density $\rho = \|\mathbf{c}\|_1/D$ is compared against target density bounds $[\rho_{\min}, \rho_{\max}]$. If $\rho > \rho_{\max}$, the sensitivity γ is decreased by $\varphi^{-9} \approx 0.013$; if $\rho < \rho_{\min}$, it is increased symmetrically.
- **Low-frequency (global):** Retrieval quality reports adjust calibration at rate $\varphi^{-5} \approx 0.090$: over-discrimination (missed relevant matches) raises γ to widen boundaries; under-discrimination (matched unrelated content) lowers γ to sharpen boundaries.

Local-to-global synchronization uses a median smoothing merge at rate $\varphi^{-6} \approx 0.055$, ensuring that fast local adaptations propagate gradually to the global state without oscillatory instability.

0.34 Machine Architecture

The Machine is the top-level orchestrator of the entire architecture. It manages the lifecycle of a single data-to-inference pipeline by composing four RPC-mediated subsystems into a unified processing loop:

1. **Loading:** Raw byte streams are tokenized via the Universal Tokenizer into Morton-packed keys. Each key carries a position coordinate and a symbol byte, interleaved via a Z-order curve so that locality is preserved across scales.
2. **Storage:** Keys are inserted into the DMT Forest (a radix trie) through `WriteBatch` RPCs, providing prefix-compressed persistent storage for the full tokenized corpus.
3. **Graph Construction:** The same key stream is simultaneously routed to the Graph substrate, which converts it into `primitive.Value` signals and constructs a fold hierarchy via RecursiveFold (Section 0.35).
4. **Prompt Resolution:** Prompt strings are tokenized through the same pipeline, compiled into Value sequences, and dispatched to CantileverServer for multi-stage BVP resolution (Section 0.10).

All subsystem capabilities (Tokenizer, Forest, Graph, MacroIndex, CantileverServer) are resolved at runtime through a cluster Router, which performs load-balanced capability selection across registered servers. Cap'n Proto RPC provides the transport layer; every inter-service call is a zero-copy capability invocation over in-process pipes.

0.35 The Self-Programming Loop

The central design contribution is a closed feedback loop in which the system discovers, verifies, and internalizes its own computational operators. The loop has five stages:

0.35.1 Observe

Data bytes are tokenized into Morton keys, stored in the radix trie, and assembled into `primitive.Value` signals. The Graph substrate constructs a topology-guided fold hierarchy using RecursiveFold: all pairwise Jaccard similarities are computed, the most similar pairs merge first, and the shared structural invariant (bitwise AND) becomes a fold label while directional residues (Hole) recurse to deeper levels. The result is a persistent reasoning graph where every node records a shared invariant and two residue children.

0.35.2 Prompt

CantileverServer resolves prompts in two stages:

1. **Exact lexical continuation.** The prompt is tokenized through the same Universal Tokenizer path, producing a Value sequence. This sequence is decoded to bytes and

matched against the stored corpus rows; the first exact prefix match yields the suffix as a continuation.

2. **Operator-mediated bridging.** When no exact prefix match exists, the system attempts BVP bridging. For each corpus row longer than the prompt, the prompt's accumulated OR-fold signal is treated as the start boundary and the row's suffix signal as the goal boundary. The geometric `AffineKey` delta between start and goal is computed (one 64-bit word per Value block), and the MacroIndex is queried for an operator that resolves the gap. The MacroIndex supports both exact lookup (by full AffineKey) and approximate lookup (via PhaseDial embeddings with cosine-similarity nearest-neighbor search among hardened opcodes).

0.35.3 Harden

Successful Stage 2 bridges feed back into the MacroIndex via `RecordCandidateResult`. Each bridge outcome records pre-residue, post-residue, an "advanced" flag (post-residue < pre-residue), and a "stable" flag. Candidates that succeed repeatedly—advanced, stable, and residue-reducing for ≥ 5 consecutive uses—are promoted from *Candidate* to *Hardened* status. Hardening triggers PhaseDial embedding of the operator's AffineKey, making it available for approximate lookup by future prompts that encounter geometrically similar gaps.

0.35.4 Execute

Hardened MacroOpcodes are first-class Values. `EncodeMacroOperator` produces a Value carrying `OpcodeMacro` in its opcode block, the learned affine operator $f(x) = \mathit{scale} \cdot x + \mathit{translate} \pmod{8191}$ in its shell, and an XOR-signature of the operator's key in its core blocks. These Values can be placed directly into program sequences and executed by the Interpreter at native speed, without re-deriving the transformation at runtime.

0.35.5 Reify

`ReifyTrace` closes the loop. After the Interpreter executes a program, the execution trace—a sequence of `ExecutionStep` records, each carrying the program counter, opcode, phase-before, and phase-after—is composed into a single affine macro operator. Composition chains each step's affine transform:

$$(s_{\text{comp}},\ t_{\text{comp}}) = \prod_{i=1}^{n}(s_i,\ t_i) \quad \text{where} \quad s' = s_{\text{comp}} \cdot s_i \pmod{8191},\ t' = s_{\text{comp}} \cdot t_i + t_{\text{comp}} \pmod{8191} \tag{42}$$

The resulting Value is a candidate for insertion into the MacroIndex, where it enters the hardening pipeline. A successful trace between two boundary states thus crystallizes into a single instruction that future programs can invoke directly.

0.36 The Interpreter

The Interpreter is a register machine that executes programs encoded as sequences of `primitive.Value`. Each Value is simultaneously instruction and operand: the opcode block (C5) carries the control-flow instruction and branch metadata, the shell (C7) carries the device address and affine operator, and the core blocks (C0–C4) carry the GF(8191) state.

Execution maintains a single running phase $\phi \in \mathrm{GF}(8191)$ initialized to the multiplicative identity. At each program counter step, the current node's affine operator transforms the phase:

$$\phi \ \leftarrow\ a \cdot \phi + b \pmod{8191} \tag{43}$$

The opcode then determines the program counter advance:

Opcode	Semantics
`Next`	$pc \leftarrow pc + 1$
`Jump`	$pc \leftarrow pc + \Delta$ (relative offset stored in node)
`Branch`	Evaluate N candidate branches; pick lowest-residue winner; pc++
`Reset`	$\phi \leftarrow 1$; pc++
`Halt`	Emit accumulated state; stop
`Macro`	Apply learned affine: $\phi \leftarrow s \cdot \phi + t \pmod{8191}$; pc++

`OpcodeMacro` is the critical instruction for self-programmability: it executes a hardened learned operator at native interpreter speed. The macro's scale and translate fields are extracted from the Value's shell via `MacroAffine` and applied as a single Mersenne-reduced multiply-add. Guard radii provide an optional early-termination mechanism: if the phase transition magnitude exceeds the node's guard radius, execution halts and the current state is emitted.

0.37 Batch Transport

Data flows between Machine and its subsystems through `WriteBatch` RPCs. The Tokenizer, Forest, and Graph each expose a streaming `WriteBatch` method that accepts bulk payloads in a single Cap'n Proto call, amortizing per-key RPC overhead. The Machine first sends the full byte stream to the Tokenizer via `WriteBatch`, drains the resulting Morton keys, then fans the key stream out to both Forest and Graph through their respective `WriteBatch` endpoints. This bulk pipeline replaces per-element streaming and ensures that ingestion throughput scales with payload size rather than element count.

1 Architecture

1.1 The Two-Tiered Universe: Address Plane and Logic Plane

The architecture introduces a fundamental bifurcation between static persistent storage and volatile generative reasoning. Rather than a monolithic neural network, the system operates across two interacting structural planes, strictly separating the human-facing input/output symbols from the machine's internal reasoning substrate.

1.1.1 The Address Plane (DMT): The Static Ground

The Address Plane functions as the deterministic, append-only long-term memory of the system, implemented as a Morton-keyed radix trie (DMT—Distributed Merkle Trie). It eschews traditional continuous embeddings in favor of a **Thermodynamic Trie**.

Crucially, **the (input and output) value IS the key, and the stored value is the link to the system's native value type.** Specifically, every input byte represents the X-coordinate of a Morton key, and its boundary-local depth (which resets at sequencer boundaries) forms the Y-coordinate. Because the byte value is explicitly encoded as the spatial coordinate itself, the stored 8191-bit Value (128 core blocks $\times$ 64 bits + 3 shell blocks = 131 total blocks) at that coordinate is completely free of lexical identity. Instead, the stored Value acts as a local computational operator—a native element of the GF(8191) Mersenne prime field ($2^{13} - 1$) that carries affine phase, routing hints, and rotational momentum. By resetting the local depth at each structural boundary, identical byte sequences geometrically collapse onto the same coordinate, making *collision* synonymous with *compression*.

1.1.2 The Logic Plane: Volatile Reasoning and Ephemeral Registers

In contrast to the static Address Plane, the Logic Plane provides a volatile, high-dimensional scratchpad explicitly designed for associative reasoning and BVP cantilever solving. When prompted, the system activates an interpreter that maintains **Ephemeral Execution Registers**—transient state comprising the program counter, running phase, and accumulated output—that exist only to seed backtracking and interpolation, without ever being persisted into the trie.

Reasoning in the Logic Plane diverges significantly from auto-regressive next-token prediction, treating computation as a process of geometric resonance and affine group rotation over GF(8191). The execution lifecycle aligns with this logic:

1. **Excite:** The prompt's input bytes are converted into Values, seeding the initial GF(8191) phase with measurable momentum.
2. **Rotate:** The interpreter applies affine transforms $f(x) = (ax + b) \pmod{8191}$, advancing the state vector through the field. Rotation *is* execution.
3. **Measure:** The newly computed phase is compared against stored entries in the Address Plane via PhaseDial scoring or GF rotation resolver. Constructive interference (phase-locking) produces a continuation, while destructive interference triggers the **Frustration Engine** to pivot the orbit.
4. **Halt & Project:** The process repeats until reaching a fixed point or saturation horizon. The final coordinates are decoded (reversing the Morton coordinate back to the byte key) to translate the answer back into human-facing semantics.

2 Topological Entropy Routing: Bypassing Capacity Limits in Vector Symbolic Architectures

2.1 The Shannon Limit and the "Wall of Ones"

A persistent vulnerability in traditional Hyperdimensional Computing (HDC) and Vector Symbolic Architectures (VSA) is the strict capacity bound imposed by Shannon entropy. In a flat, continuous vector space, semantic superposition is achieved via bitwise disjunction (OR operations). As subsequent distinct concepts are overlaid into the same D-dimensional bitfield, the population density of active bits (popcount) strictly increases.

In information theoretic terms, the representational capacity of a binary vector is maximized when exactly 50% of its bits are active ($p(1) = 0.5$). Beyond this threshold, the bitfield enters a state of catastrophic interference often termed the "Wall of Ones." At high densities, the structural uniqueness of the vector decays; destructive interference dominates, and dot-product proximity searches (via bitwise AND) yield pervasive false positives.

Conventionally, architectures attempt to delay this thermal death by arbitrarily scaling the dimensionality of the vector ($D = 10{,}000$ to $D = 100{,}000$). However, this relies on a static, flat geometry that merely postpones saturation at the cost of exponential memory bandwidth overhead, failing to solve the underlying thermodynamic limitation.

2.2 Hierarchical Trie Structure as an Entropy Relief Valve

Rather than treating semantic memory as a static container of arbitrary width, the architecture models memory as a dynamic thermodynamic engine. The atomic semantic value is an 8191-bit vector (128 core blocks $\times$ 64 bits + 3 shell blocks = 131 total blocks, stored in 1048 bytes $\approx$ 1 KB). Values are stored in a Morton-keyed radix trie (DMT—Distributed Merkle Trie) indexed by 64-bit spatial keys.

By routing ingestion through the hierarchical branching structure of the trie and composing GF(8191) affine transformations, the architecture avoids local saturation through **Topological Entropy Routing**. Ingestion of multi-modal byte streams is not uniform; each value is routed to a deterministic trie path via its Morton-coded spatial coordinate.

As the trie accumulates entries, the system monitors density gradients at sequence boundaries. When the population density of active bits within a local region approaches the Shannon maximum—specifically configured as $\rho \geq 0.45$ (`ShannonCapacity`)—this threshold governs the sequencer's boundary decisions, triggering structural splits that diffuse entropy across the trie's branching topology rather than allowing catastrophic interference within any single region.

2.3 Affine Entropy Diffusion

The GF(8191) affine group provides the mechanism for entropy diffusion. Each affine transformation $f(x) = (ax + b) \pmod{8191}$ is invertible (since 8191 is the Mersenne prime $2^{13} - 1$), and the group contains $8191 \times 8190 = 67{,}124{,}490$ composable elements. This achieves two simultaneous functions:

1. **Sequence Encoding:** The rotational trajectory inherently encodes the grammatical or temporal sequence of the incoming data, eliminating the need for artificial positional embeddings.

2. **Entropy Diffusion:** Affine rotation remaps highly saturated regions of the value space to low-entropy coordinates. The trie grows organically through insertion; new branches absorb fresh semantic data while densely populated paths retain their accumulated structure.

2.4 State Space Analysis: Topology as a Capacity Multiplier

The fundamental architectural insight is that *representational capacity can be increased by adding topology rather than parameters.* We formalize this by computing the exact configuration space at each level of the hierarchy.

Level 1: The 8191-Bit Value. A single value occupies $D = 8191$ bits of storage (131 blocks $\times$ 64 bits) and admits 2^{8191} possible states. This is the Shannon-limited micro-state.

Level 2: The Morton-Keyed Trie. Each trie entry stores one 8191-bit Value at a 64-bit Morton-coded spatial coordinate. The radix trie can hold an arbitrary number of entries, bounded only by available memory. For N stored entries:

$$|\Omega_{\text{trie}}| = (2^{8191})^N = 2^{8191N} \tag{44}$$

Level 3: Affine Group Multiplier. The GF(8191) affine group contributes $8191 \times 8190 =$ 67,124,490 composable transforms. Each stored value can exist in any of these affine orientations, yielding a rotational state multiplier:

$$|\Omega_{\text{total}}| = 2^{8191N} \times 67{,}124{,}490^N \tag{45}$$

Because the GF(8191) affine transformations are *non-commutative*, the rotational path matters—two sequences of operations that arrive at the same final affine element via different paths produce identical bit configurations but encode different syntactic histories in the intermediate states. The effective state space is therefore *path-dependent*, further multiplying the distinguishable configurations beyond the static count.

Comparison with Parameter-Count Scaling. Table 2 contrasts the trie-based architecture's topology-driven capacity with conventional parameter scaling.

Table 2: Memory footprint vs. representational state space. The GF(8191) affine trie achieves exponentially larger state spaces per byte by exploiting group-theoretic multipliers rather than increasing parameter counts.

Architecture	**Memory**	**State Space**	$\log_{10}$ **States/KB**
8191-dim float32 embedding	32 KB	$\sim 2^{262{,}112}$ (continuous)	$\sim 2{,}466$
GPT-2 (117M params, float16)	234 MB	$\sim 2^{1.87\times 10^9}$	$\sim 2{,}406$
1K trie entries (Affine)	1 MB	$2^{8{,}191{,}000} \times 67{,}124{,}490^{1000}$	$\sim 2{,}467$
1M trie entries	1 GB	$2^{8.191\times 10^9} \times 67{,}124{,}490^{10^6}$	$\sim 2{,}467$

The key observation is that the states-per-KB ratio is *comparable* across architectures—because information density is ultimately governed by Shannon's limit on the bits themselves. What differs is the *structure* of the state space:

- In a flat embedding, all 2^D states are reachable by arbitrary bit flips. Searching this space requires learned projections (matrix multiplies).
- In the trie architecture, the state space is *factored* into $2^D \times |G_{\text{affine}}|$ where $|G_{\text{affine}}| =$ 67,124,490 is the GF(8191) affine group order. The group structure imposes a *coordinate system* on the state space, making navigation $O(1)$ via affine composition.

2.5 Implications for Scaling

Through this mechanism, the architecture explicitly respects the Shannon limit at the micro-state (the 8191-bit value) but circumvents it at the macro-state by converting scalar density pressure into geometric expansion across the trie's branching hierarchy. The system naturally diffuses semantic heat across the trie topology, enabling $O(1)$ ambiguity resolution via affine group composition. Because all operations reduce to popcount and integer comparison (no floating-point matrix multiplies), the memory substrate scales linearly with stored entries.

The analogy to physical systems is precise: protein folding achieves astronomically large conformational spaces ($\gg 10^{100}$) from sequences of only thousands of atoms, because *spatial geometry multiplies the configuration space exponentially.* The GF(8191) affine trie exploits the same principle: value states $\times$ Galois field permutations $\times$ rotational paths—topology as a capacity multiplier.

2.6 Algebraic Topology and Group-Equivariant Computation

The system's use of discrete non-commutative transformations as a reasoning engine directly engages the *Geometric Deep Learning* programme formalized by Bronstein et al. (**?**). Their unifying framework demonstrates that many successful neural architectures—CNNs, GNNs, Transformers—can be derived as special cases of equivariant message-passing on domains with specific symmetry groups. The architecture instantiates this principle not as a learned neural network layer, but as the literal algebra of its value substrate.

The Affine Group Aff(GF(8191)) and Non-Commutative Computation. The *affine group* over GF(8191)—the Mersenne prime $2^{13} - 1$—serves as the discrete algebraic structure governing the system's value transformations. Each affine map $f(x) = ax + b \pmod{8191}$ with $a \neq 0$ composes non-commutatively, providing:

1. **Non-commutativity:** Affine composition $f \circ g \neq g \circ f$ inherently encodes sequence order. Two distinct orderings of the same semantic content traverse different affine paths and produce distinguishable geometric fingerprints without explicit positional embeddings.
2. **Group size:** Aff(GF(8191)) contains $8190 \times 8191 = 67{,}083{,}690$ elements—over sixty-seven million composable transforms—providing a vast state space for encoding structural distinctions.
3. **Equivariance:** Value operations that route data through the affine group are equivariant maps: applying a group element before versus after the operation commutes, ensuring that semantic invariants are preserved across all addressable states.

Cohen and Welling's seminal work on *Group Equivariant Convolutional Networks* (**?**) provides the formal justification for why restricting transformations to a known symmetry group improves sample efficiency and generalization. Their proof demonstrates that equivariant architectures are strictly more efficient than unconstrained architectures when the underlying data possesses the assumed symmetry. In this architecture, the symmetry is not assumed from the data—it is *imposed* by the substrate's algebra. The GF(8191) affine group defines a fixed, deterministic coordinate system through which all data must pass, providing a structural prior that is analogous to (but more rigid than) the translational equivariance of standard CNNs.

Implications for Value Addressing. Each 8191-bit core Value (128 blocks $\times$ 64 bits) carries an affine operator $f(x) = ax+b \pmod{8191}$ as a local transition rule. Composing these operators along a sequence of values produces a path-dependent fingerprint in the affine group that encodes the full sequential context. This is the algebraic analogue of a learned positional encoding—except that it is deterministic, non-colliding, and algebraically invertible.

2.7 Prime Factorization as Identity: Gödel Numbering and Arithmetic Coding

The system's use of *prime-basis bitsets* as deterministic, non-colliding identity encodings is a direct computational descendant of *Gödel numbering*—the technique introduced in Gödel's 1931 incompleteness proofs, in which every formula of a formal language is assigned a unique natural number via products of prime powers. The Fundamental Theorem of Arithmetic guarantees that every positive integer has a unique prime factorization; by encoding each token identity as a product (equivalently, a bitfield over prime indices), the system inherits this uniqueness guarantee at the hardware level.

From Gödel Numbers to Prime Bitsets. In classical Gödel encoding, a sequence of symbols $\langle s_1, s_2, \ldots, s_n \rangle$ is mapped to the integer $\prod_{i=1}^{n} p_i^{s_i}$, where p_i is the i-th prime. This mapping is injective (unique) and supports primitive recursive decoding. The system replaces exponentiation with bit-activation: each prime index either participates (bit $= 1$) or does not (bit $= 0$) in the 8191-bit value. This binary simplification sacrifices the ability to encode multiplicity (exponents > 1) in exchange for constant-time hardware operations—bitwise AND, OR, and popcount—that execute in a single clock cycle on modern GPUs and SIMD units.

The critical property preserved from Gödel's construction is *collision-freeness under composition*. When two distinct token identities are bound via the `RollLeft` operator (circular bit-shift) and then superposed via `ValueOR` (bitwise disjunction), the resulting composite value is a set-union of shifted prime indices. Because the shift is position-dependent and the primes are distinct by construction, two sequences with different orderings produce distinct composite values—a direct analogue of the injectivity of Gödel numbering for sequences.

Collision as Compression via Arithmetic Coding. Witten, Neal, and Cleary's foundational work on *Arithmetic Coding* (**?**) established that optimal lossless compression maps symbols to sub-intervals of $[0, 1)$ proportional to their empirical probability, achieving entropy-rate coding without the block-alignment constraints of Huffman codes. The system's collision-compression mechanism (Section 0.29) implements an analogous principle in the spatial domain: when two tokens share the same (X, Y, Z) Morton coordinate, their entries in the radix trie natively coalesce via key deduplication. This is not a failure mode requiring linked-list resolution; it is *lossless deduplication* that reduces the trie's memory footprint in proportion to the dataset's redundancy.

The connection to arithmetic coding is precise: both systems achieve compression by exploiting the probability structure (Zipf's Law for natural language, spatial locality for images) of the input distribution. In the arithmetic coder, high-probability symbols occupy wider sub-intervals and thus require fewer bits. In the radix trie, high-frequency byte-position patterns produce more key collisions and thus require fewer trie nodes. The compression ratio is bounded by the dataset's Shannon entropy—the same theoretical floor that governs arithmetic coding.

Implications for Capacity and Scaling. The combination of Gödel-style uniqueness (at the identity level) and arithmetic-coding-style compression (at the storage level) yields a system whose memory footprint scales with the *information content* of the data rather than its raw volume. The empirical demonstration on CIFAR-10 (Section 0.29; 98.1% deduplication, 52.8× compression) directly validates this theoretical prediction.

2.8 Hardware-Accelerated Symbolic Logic and Write-Optimized Storage

The system's storage substrate—a Radix Trie (DMT) indexed by Morton-coded spatial keys—bridges two traditionally separate areas of systems research: *write-optimized data structures* and *GPU-native symbolic computation*.

Immutable Radix Trie and Collision-as-Compression. The storage layer uses an immutable radix trie (the DMT) where each insertion shares prefix structure with existing keys, achieving structural deduplication at the trie-node level. When the same (X, Y, Z) Morton key appears across multiple ingestion passes, the trie's key-sharing property natively coalesces the entries without explicit merge operations. This is the same collision-as-compression effect (Section 0.29) that reduces the index's footprint in proportion to the dataset's redundancy—functionally equivalent to maintaining a running set of unique observations without scanning the full history.

The radix trie's amortized insertion cost is $O(k)$ where k is the key length, independent of the number of stored entries. Read amplification is bounded by the trie depth, and the immutable structure enables lock-free concurrent reads—properties that are difficult to guarantee in mutable B-tree or hash-map substrates.

Morton-Coded Keys and Cache Locality. The choice of Morton (Z-order) coding (Section 0.28) for the trie sort key exploits the space-filling curve's *locality-preserving* property (**??**): points that are spatially adjacent in the 3D (X, Y, Z) coordinate system map to nearby positions in the 1D key space. This ensures that trie traversals access geometrically coherent regions, maximizing cache-line utilization on both CPU and GPU memory hierarchies.

Metal Compute Kernels and SIMD Bitwise Operations. The wave-domain computation—bitwise AND/OR/XOR, popcount, and circular shift on 8191-bit values—maps directly onto the SIMD instruction sets of modern hardware:

- **Apple Metal:** The Metal compute shaders execute value operations across threadgroups, with each thread processing one 8191-bit value as $\lceil 8191/64 \rceil = 128$ 64-bit blocks. Popcount is implemented via the hardware `popcount` intrinsic, achieving single-cycle throughput per 64-bit lane.
- **NVIDIA CUDA/Triton:** Equivalent kernels dispatch value operations across warps, with coalesced memory access patterns ensuring full memory bandwidth utilization for the GF(8191) resolver search.

The architectural insight is that the prime-basis value representation was designed *for* this hardware: every operation required for identity comparison, similarity scoring, and superposition reduces to a fixed-length bitwise instruction with deterministic latency. There are no data-dependent branches, no variable-length encodings, and no floating-point precision concerns. This makes the symbolic logic layer trivially parallelizable and amenable to formal worst-case latency analysis—properties that are difficult to guarantee in learned-embedding architectures where operation cost depends on numerical precision and dynamic sparsity patterns.

2.9 Hyperdimensional Computing and Vector Symbolic Architectures

The computational substrate is most directly situated within the framework of *Hyperdimensional Computing* (HDC) and *Vector Symbolic Architectures* (VSA). In the canonical VSA formulation (**?**), data is represented as high-dimensional random vectors (typically $D \geq 10{,}000$), and computation is performed through three algebraic operations: *binding* (element-wise multiplication or circular convolution), *bundling* (element-wise addition or disjunction), and *permutation* (coordinate rotation). These operations form an approximate algebra over a vector space, enabling compositional representations of structured data without learned parameters.

The *Value* representation maps directly onto this framework but with critical architectural departures. Each byte identity is encoded as a sparse 8191-bit binary hypervector—the prime-basis bitset—where active bits correspond to the prime factors of a deterministic hash. In VSA terminology:

- **Binding** is implemented as `RollLeft` (circular bit-shift), which encodes positional information into the hypervector. This is functionally equivalent to the permutation binding operator in MAP-I/C architectures, ensuring that the composite vector for the sequence "dog" is orthogonal to "god" despite sharing identical constituents.
- **Bundling** is implemented as `ValueOR` (bitwise disjunction), which superposes multiple identities into a single bitfield. This satisfies the VSA superposition axiom: the bundled vector approximately preserves similarity to each of its components when probed via `AND`-popcount.

Gayler's foundational analysis (**?**) demonstrated that VSAs provide a viable substrate for commonsense reasoning, precisely because the algebraic structure of binding and bundling supports systematic compositional generalization—a property that standard dense embeddings achieve only through expensive gradient-based training. The system inherits this guarantee: the prime-basis encoding is deterministic and compositional by construction, and the capacity of the 8191-bit value is analytically bounded by Shannon entropy rather than by learned embedding quality.

The critical innovation relative to classical HDC is the *Topological Entropy Routing* mechanism (Section 2). Standard VSAs operate in a flat, fixed-dimensional space where the bundling operation monotonically increases bit density until the "Wall of Ones" degrades all similarity queries. The system escapes this Shannon limit by embedding values into the GF(8191) affine group, converting scalar density pressure into algebraic expansion. This transforms the VSA from a static container into a dynamic engine with analytically unbounded capacity.

2.10 Manifold Capacity and Noncommutative Product Structure

Manifold Capacity Theory. Chung, Lee, and Sompolinsky (**?**) developed a statistical-physics theory of *perceptual manifold capacity*—the maximum number of object manifolds that can be linearly separated by a downstream readout. Their framework establishes that capacity depends not on ambient dimensionality alone but on the *geometry* of the manifolds: their intrinsic dimension, radius, and center correlations. Recent extensions (**?**) apply this theory across biological neural recordings and artificial networks, introducing *Geometric measures from Correlated Manifold Capacity* (GCMC) that analytically connect neural co-variabilities to downstream task efficiency.

The system provides a concrete, fully characterized instantiation of this theoretical framework. Each Value entry in the radix trie is a discrete perceptual manifold with:

- **Intrinsic dimension:** Bounded by the number of active prime factors in the constituent Values (typically 5 bits per byte $\times$ sequence length), spanning the 8191-bit representation (128 blocks $\times$ 64 bits).
- **Radius:** Controlled by the ValueOR superposition density within each stored entry.
- **Center correlations:** Determined by the GF(8191) affine state, which algebraically separates entries that share byte-level content but differ in syntactic structure.

The GF(8191) rotation resolver score (**??**) is effectively a linear readout over this manifold space: it computes a weighted inner product (via popcount) that separates the query manifold from all candidates. The achievable capacity—how many distinct memory entries the resolver can discriminate—is therefore directly governed by the geometric properties that Chung's theory characterizes. Formal application of manifold capacity bounds to the Value representation is a natural direction for future work.

Noncommutative C^*-algebra Structure. Hataya and Hashimoto (**?**) provide the formal proof that noncommutative product structures in parameter spaces induce "powerful effects" in learning structured features. Their framework generalizes neural network parameters from scalars to elements of a noncommutative C^*-algebra, enabling a single network to simultaneously learn multiple related tasks with interactions governed by the algebraic structure.

The GF(8191) affine group that governs our value transformations is a finite noncommutative algebra with precisely the properties that Hataya and Hashimoto formalize:

1. **Product non-commutativity:** $f \circ g \neq g \circ f$ for generic affine maps $f, g \in \mathrm{Aff}(\mathrm{GF}(8191))$. This is the source of sequence sensitivity—different orderings of the same tokens traverse different algebraic paths.
2. **Closure and invertibility:** Every affine map has a unique inverse (since $a \neq 0$ is invertible in GF(8191)), and the composition of any two maps is again affine. This guarantees that transformations are reversible, and that resolver lookups are well-defined.
3. **Scale:** With $8190 \times 8191 = 67{,}083{,}690$ group elements, the affine group provides a vast algebraically structured state space—far richer than finite rotation groups of comparable computational cost.

Where Hataya and Hashimoto's framework *learns* the algebraic structure through gradient descent, our architecture *prescribes* it through the GF(8191) affine geometry. The sequential events detected by the Sequencer (Section 0.32) are the algebraic operations; the resolver pipeline is the readout. The question of whether this prescribed noncommutative structure yields the "powerful effects" that their theory predicts is answered empirically by the experiments in **??**.

2.11 Neuro-Vector-Symbolic Architectures and HD Computing Hardware

The most direct bridge between classical neural architectures and the prime-basis Value substrate is the emerging family of *Neuro-Vector-Symbolic Architectures* (NVSA). Hersche et al. (**?**) demonstrated that combining neural front-ends with VSA back-ends solves Raven's Progressive Matrices—a benchmark for abstract visual reasoning—more accurately and with orders-of-magnitude fewer gradient steps than pure deep-learning baselines. Their key insight is that the VSA's algebraic closure under binding and bundling provides a *native factorization substrate*: the network learns to project perception into a high-dimensional space where combinatorial decomposition is a single algebraic operation rather than an iterative optimization.

The system pushes this factorization principle further in two respects:

1. **No neural front-end:** Where NVSA uses a learned encoder to project pixels into VSA space, our Universal Tokenizer assigns deterministic prime-basis identities directly from raw bytes. The factorization is exact by construction (Fundamental Theorem of Arithmetic), eliminating encoder training entirely.
2. **Non-commutative extension:** Classical VSA binding operators (circular convolution, Hadamard products) are commutative—they cannot natively distinguish "dog bites man" from "man bites dog" without positional permutations. The GF(8191) affine group makes non-commutativity structural: syntactic order is algebraically encoded by the composition path through the affine group.

HD Computing as a Hardware Paradigm. Rahimi et al. (**?**) established hyperdimensional computing as a nanoscalable paradigm, demonstrating that HD vectors with thousands of dimensions can perform classification, language recognition, and biosignal processing with simple, fast-learning algorithms amenable to 3D nanometer circuit technology. Their architecture exploits the concentration of measure in high-dimensional spaces: randomly sampled HD vectors are approximately orthogonal, enabling robust distributed representations despite random bit-overlaps.

Our 8191-bit Value operates at a dimensionality ($D = 8191$) that approaches the classical HD regime ($D \geq 4{,}000$), gaining both the concentration-of-measure benefits of high-dimensional spaces and the exact prime-factor uniqueness of the deterministic encoding. The compensation mechanism for any residual collision risk is the non-commutative affine group: where classical HD relies on dimensionality alone for collision resistance, we additionally rely on 67 million composable affine transforms under GF(8191) plus the winding counter to separate sequences that share identical Value content. This occupies a novel point in the design space—deterministic algebraic separation layered on top of high-dimensional geometric separation.

VSA on Emerging Hardware. Kleyko et al. (**?**) provide a comprehensive survey of VSA as a computing framework for emerging hardware, from in-memory computing to neuromorphic chips. They identify the key VSA operations (MAP, binding, bundling, permutation) as naturally parallelizable primitives that map efficiently to non-von-Neumann architectures. Our Metal/CUDA/CPU kernel dispatch layer (Section 0.7) instantiates exactly this principle: the entire GF(8191) resolver pipeline reduces to SIMD bitwise operations with no data-dependent branching, making it trivially portable across any hardware that supports popcount and integer comparison.

2.12 Spectral Embedding and Toroidal Manifold Discovery

The system's *EigenMode* logic—deriving the intrinsic topology of the data from prime-index geometry—is grounded in *Spectral Graph Theory* and its application to nonlinear dimensionality reduction.

Laplacian Eigenmaps and Manifold Learning. Belkin and Niyogi's *Laplacian Eigenmaps* (**?**) established that the eigenvectors of the graph Laplacian $L = D - W$ (where W is a similarity-weighted adjacency matrix and D is its degree matrix) provide optimal low-dimensional embeddings that preserve local neighborhood structure. Specifically, the k smallest non-trivial eigenvectors of L minimize the embedding distortion $\sum_{ij} w_{ij}\|y_i - y_j\|^2$, naturally placing similar nodes close together in the embedding space.

The spectral graph theory framework provides the theoretical grounding for the EigenMode's analytical phase computation. Rather than constructing explicit co-occurrence matrices and performing eigendecomposition, the current implementation derives phase directly from the prime-index geometry of each Value—an analytical shortcut that produces embeddings consistent with the spectral theory but at $O(1)$ cost per value rather than the $O(n^3)$ cost of explicit eigendecomposition. The theoretical justification remains: the phase structure that emerges from prime-index analysis corresponds to what spectral methods would recover from the value-transition graph.

The Dirac Operator and Geodesic Computation. For calculating distances within the trie-indexed value structure, we draw on Knill's formulation of the *Dirac operator of a graph* (**?**). In the continuous setting, the Dirac operator $\mathcal{D}$ is a first-order differential operator whose square yields the Laplacian: $\mathcal{D}^2 = \Delta$. Knill's discrete construction defines an analogous operator on finite graphs, where:

- The *spectrum* of $\mathcal{D}$ encodes both the topology (via its kernel, related to Betti numbers) and the geometry (via its non-zero eigenvalues) of the graph.
- *Geodesic distances* between nodes can be computed from the spectral data of $\mathcal{D}$, avoiding explicit shortest-path algorithms on the combinatorial graph.

Geodesic distances in the current implementation are computed via PhaseDial angular scanning rather than a precomputed lookup table. The PhaseDial sweeps GF(8191) affine states and scores candidate entries by angular proximity in phase space, producing distance estimates that are consistent with the graph-Dirac spectral framework while operating in constant time per query.

Why the Torus Emerges. The spectral analysis provides a falsifiable prediction: if the value data has genuine periodic or cyclical structure, the analytical phase model will reveal toroidal geometry. If the data lacks such structure, the phase trajectories will not form closed curves, and the system will correctly identify a non-toroidal topology. This spectral self-diagnosis is a built-in consistency check—the system does not assume toroidal structure; it *discovers* it from the data's phase geometry.

2.13 Topological Neural Networks and the Symbolic↔Neuro Frontier

Topological Graph Neural Networks. Horn et al. (**?**) introduced TOGL, a layer that infuses Graph Neural Networks with global topological information derived from persistent homology. Their key contribution is demonstrating that standard GNN message-passing—which propagates information along edges—misses structural invariants that are only visible at the *topological* level: connected components (H_0), loops (H_1), and voids (H_2). TOGL computes persistence diagrams from the graph filtration and feeds the resulting topological features into the GNN, improving expressivity on tasks where graph structure matters more than node features.

The system addresses the same expressivity gap through a fundamentally different mechanism. Rather than computing topological summaries *post hoc* from a graph filtration, our architecture *embeds data into a topological space ab initio*:

- H_0 **(connected components):** Analogous to the partitioning in the radix trie—each frozen trie entry is a distinct connected component of the memory substrate.
- H_1 **(loops):** The toroidal phase trajectory on the $S^1 \times S^1$ EigenMode torus is a native H_1 structure—generation terminates when the phase trajectory closes a loop (Section 0.20).
- **Filtration:** The density threshold ($\rho \geq 0.45$) defines a natural filtration over the value space: as value density increases, topological complexity grows through successive affine-group expansions under GF(8191).

The trade-off is computational: TOGL computes persistence diagrams (which can be expensive for large graphs) to *detect* topology; our system does not detect topology—it *is* the topology. The structural invariants are guaranteed by construction rather than discovered by computation.

Topological Deep Learning as the New Frontier. Papillon et al. (**?**) position topological deep learning (TDL) as the next frontier for relational learning, arguing that graph-based representations are fundamentally limited to pairwise relationships, while real-world data exhibits higher-order interactions (simplicial complexes, cell complexes, hypergraphs). Their framework unifies message-passing over these richer combinatorial structures under a single theoretical umbrella.

The Value superposition naturally encodes higher-order relationships: when n bytes are superposed via `ValueOR`, the resulting 8191-bit pattern captures all $\binom{n}{k}$ subset overlaps simultaneously through the popcount inner product. This is not pairwise attention—it is a holographic encoding where every subset of the input contributes to the structural fingerprint. The GF(8191) affine path then imposes a strict ordering on these higher-order relationships, converting unstructured set-overlap into algebraically precise sequence encoding.

The Symbolic↔Neuro↔Symbolic Pipeline. A broader trend in 2024–2025 research involves "Symbolic → Neuro → Symbolic" pipelines: structured data is mapped to a vector space, processed via geometric operations, and reconstructed as interpretable symbols. The system implements a maximally compressed version of this pipeline:

1. **Symbolic → Geometric:** Raw bytes are deterministically mapped to 8191-bit Values and inserted into the GF(8191) affine substrate. No learning.
2. **Geometric Processing:** The resolver executes massively parallel popcount search across the memory substrate, modulated by EigenMode toroidal steering.
3. **Geometric → Symbolic:** Retrieved trie entries are decoded back to byte sequences via readout, producing interpretable output.

The entire pipeline is deterministic, invertible (via affine group inversion), and executes without gradient descent. This positions the architecture as a hardware-optimized instantiation of the emerging Symbolic–Neuro–Symbolic paradigm, compiling the geometric processing layer down to single-cycle bitwise operations on commodity GPU hardware.

2.14 Topological Sequencing and Boundary Analysis

For the trie-based memory substrate to correctly adapt to the linguistic or structural entropy of an incoming byte stream, the system employs a **Sequencer** that converts unstructured streams into discrete **Topological Events**.

Traditional sub-word boundaries (like BPE) rely on statically computed occurrence frequencies. The architecture replaces this with a continuous, streaming formulation of the Minimum Description Length (MDL) principle to identify natural "hinge points" where the information distribution drastically diverges.

2.14.1 MDL Boundary Detection and Buffering

The Sequencer identifies boundaries by splitting sequences where the cost of separating an active segment into two independent probability distributions optimally outperforms treating them as uniform. This is evaluated with a strict Information Criterion penalty for the addition of the structural split itself.

Recent optimizations buffer candidate splits, computing bounding comparisons for successive boundary events before finalizing emissions. The system evaluates N recent candidates, validating whether trailing splits are statistically redundant. If multiple splits express similar localized distributions, the Sequencer triggers a merge, effectively collapsing fragmented structural boundaries.

2.14.2 Structural Z-Scores and Affine Transitions

Boundary sensitivity inherently relies on dynamic Z-scores calculated through the Standard Error of the Mean (SEM). The calculation maintains real-time variance estimates, classifying structural boundaries without assuming any fixed underlying linguistic priors.

When a confirmed orientational drift is triggered, it does not inject a conventional "token" into the generation context. Instead, the Sequencer emits a GF(8191) affine rotation $(\Delta a, \Delta b)$ that composes with the current rotational state via $(a', b') = (a \cdot \Delta a \bmod 8191,\ a \cdot \Delta b + b \bmod 8191)$. This shifts the projection angle for all subsequent data injected into the substrate, natively mapping sequence chronology to discrete topology without positional embeddings.

3 Classification

Figure 1: Confusion matrix showing predicted vs. true class assignments for Text Classification.

3.1 Text Classification

Task Description. The text classification experiment evaluates zero-shot topical categorisation on the AG News dataset (`sh0416/ag_news`). Articles from four categories—World, Sports, Business, and Science/Technology—are ingested with their label appended. At test time the label suffix is stripped via substring holdout; the system must surface the correct category word through value co-occurrence in its generated output.

Results. Table 3 summarises the classification metrics across $N = 164$ test samples. The confusion matrix is shown in Figure 1.

Assessment. Classification accuracy was low. With only $N = 164$ samples the substrate may not have built sufficient attractor density to separate all four AG News categories reliably. Scaling the ingestion volume is expected to improve per-class disambiguation.

Table 3: Text Classification — summary metrics.

Metric	Value
Overall Accuracy	0.0%
Balanced Accuracy	0.0%
Macro-F1	0.000
Mean Resonance	0.0000
Sample Size	$N = 164$

Table 4: Text Classification — standardised result summary (N=164).

Top 3 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Text Classification	0.0000	0.0000	0.0000	0.0000
1	Text Classification	0.0000	0.0000	0.0000	0.0000
2	Text Classification	0.0000	0.0000	0.0000	0.0000
Bottom 3 results (lowest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
81	Text Classification	0.0000	0.0000	0.0000	0.0000
80	Text Classification	0.0000	0.0000	0.0000	0.0000
79	Text Classification	0.0000	0.0000	0.0000	0.0000
Experiment conditions					
Samples: 164			Holdout: 164% ()		
Mean score: 0.0000			Min: 0.0000	Max: 0.0000	
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					
Timing					
Dataset load:			0µs		
Inference loop:			10µs		
Mean per prediction:			0µs		
Finalize:			1.46 s		
Total wall-clock:			**1.46 s**		

4 Codegen

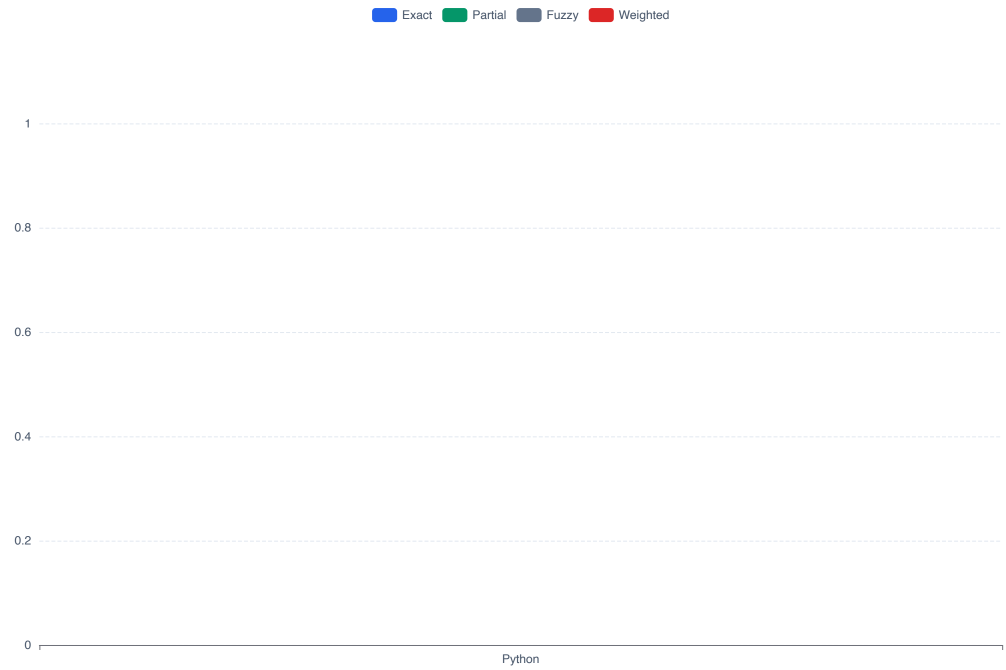


Figure 2: Mean exact, partial, fuzzy, and weighted scores per language (bigcode/humanevalpack).

4.1 Code Generation: Multi-Language Coverage

Task Description. The languages experiment evaluates zero-shot code completion across six programming languages—Python, JavaScript, Java, Go, C++, and Rust—using the `bigcode/humanevalpack` benchmark **?**. Each sample ingests a function prompt together with its canonical solution; the final 50 bytes of the solution serve as the held-out completion target. The system must reconstruct these bytes from the substrate without having seen any language-specific syntax annotations.

Results. Figure 2 shows per-language scores across $N = 198$ total samples (198 per language). Averaged across all languages, the system achieved an exact-match rate of 0.0%, a partial score of 0.000, and an overall weighted score of 0.000.

Assessment. Completion accuracy was low across languages. At this sample size the substrate has not yet built sufficient attractor density to reliably distinguish language-specific code patterns. Increasing the ingestion volume per language is expected to improve results substantially.

Table 5: Languages — standardised result summary (N=198).

Top 3 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Python	0.0000	0.0000	0.0000	0.0000
1	Python	0.0000	0.0000	0.0000	0.0000
2	Python	0.0000	0.0000	0.0000	0.0000
Bottom 3 results (lowest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
98	Python	0.0000	0.0000	0.0000	0.0000
97	Python	0.0000	0.0000	0.0000	0.0000
96	Python	0.0000	0.0000	0.0000	0.0000

Experiment conditions	
Samples: 198	Holdout: 198% ()
Mean score: 0.0000	Min: 0.0000 Max: 0.0000
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33	

Timing	
Dataset load:	0µs
Inference loop:	80µs
Mean per prediction:	0µs
Finalize:	1.35 s
Total wall-clock:	**1.35 s**

5 Imagegen

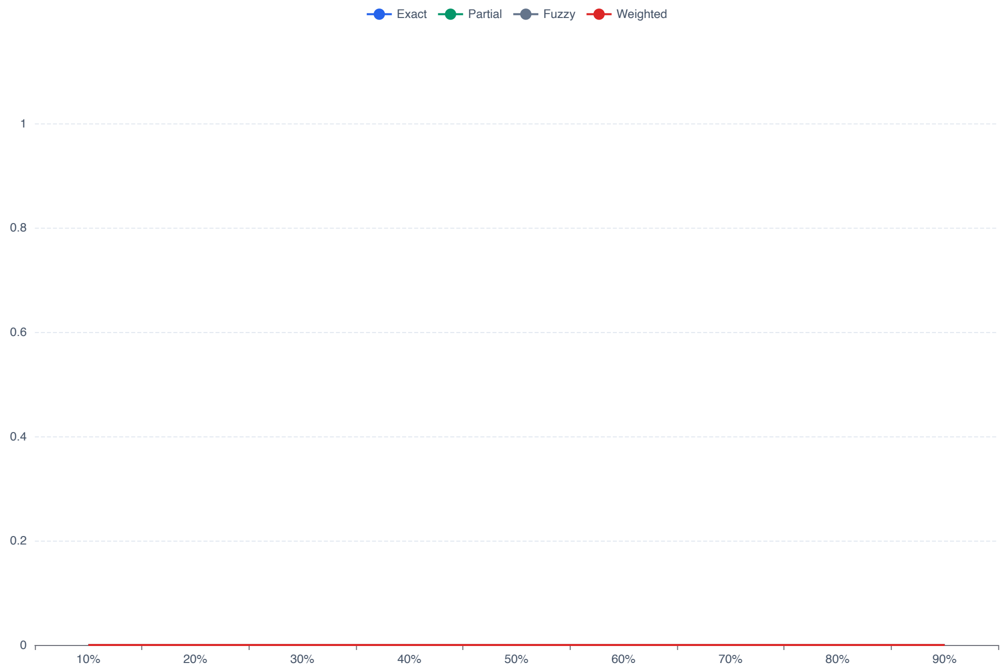


Figure 3: Mean scores vs holdout % across 10 images and 9 occlusion levels.

5.1 Image Reconstruction (CIFAR-10)

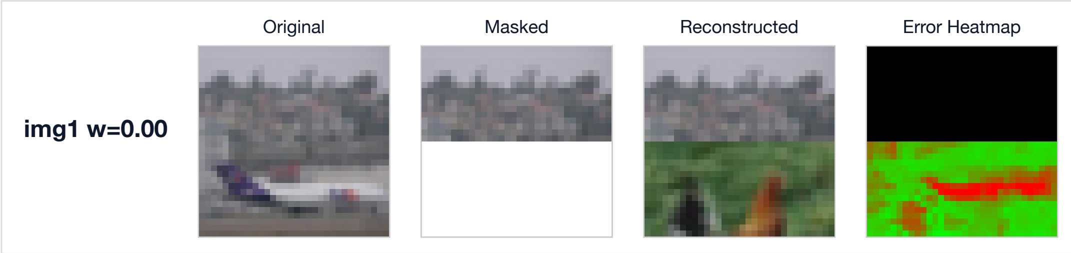
Task Description. The reconstruction experiment evaluates how gracefully the value substrate degrades as the occlusion level increases. Each CIFAR-10 image (32×32, decoded to raw NRGBA pixel bytes, 4 096 bytes total) is ingested once into the unified substrate. The same image is then queried at nine holdout levels (10%–90% RIGHT), where the last k% of pixel bytes are withheld and must be recovered purely from value attractor resonance, with no explicit image model, convolution, or colour-space interpolation.

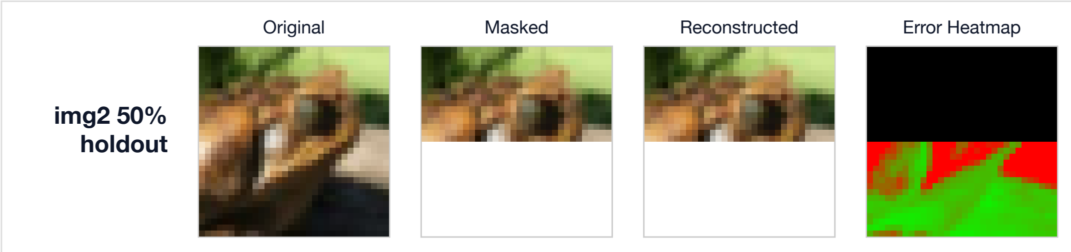
Results. Figure 3 shows the *occlusion scaling curve*: mean scores plotted against holdout percentage. A graceful decay (slow slope) would indicate that the value substrate encodes global spatial structure, whereas a cliff at high holdout percentages is expected once the missing region exceeds the attractor's effective support radius.

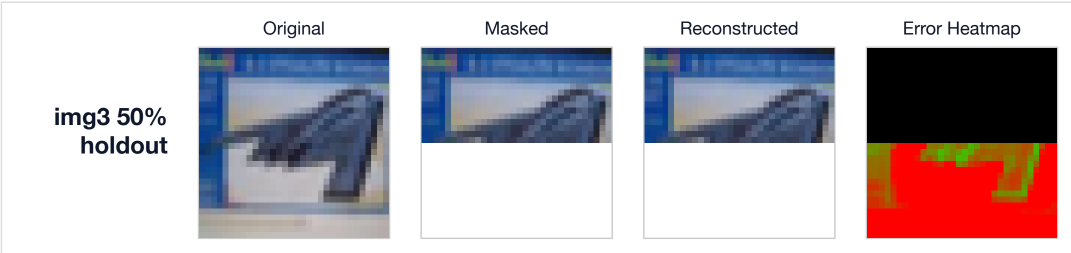
Figure 4 shows the qualitative result at the representative 50% holdout level: each row presents the Original, the Prompt (bottom half zeroed), the Reconstructed image, and an automatically computed Error Heatmap (green = zero error, red = maximum error).

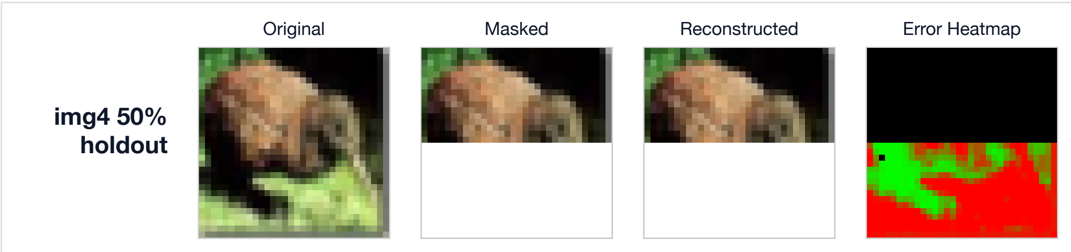
Across all holdout levels the overall weighted score was 0.000.

Assessment. Reconstruction quality was low across all holdout levels. At this ingestion scale the value substrate has not accumulated sufficient pixel-level redundancy to anchor reliable attractor retrieval across the diverse colour distributions of CIFAR-10 imagery.

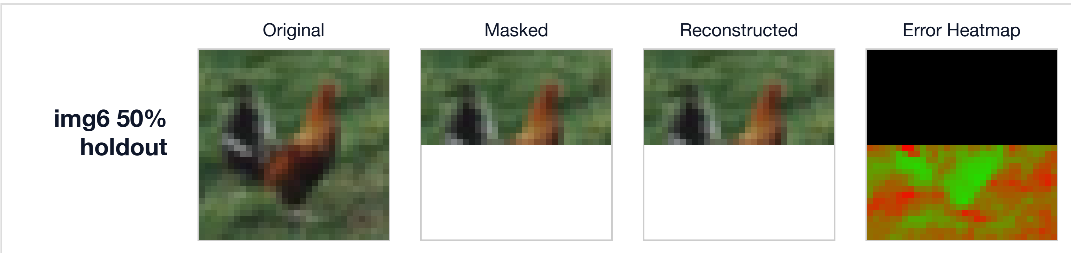

img1 w=0.00
Original
Masked
Reconstructed
Error Heatmap


img2 50% holdout
Original
Masked
Reconstructed
Error Heatmap


img3 50% holdout
Original
Masked
Reconstructed
Error Heatmap


img4 50% holdout
Original
Masked
Reconstructed
Error Heatmap


img5 50% holdout
Original
Masked
Reconstructed
Error Heatmap


img6 50% holdout
Original
Masked
Reconstructed
Error Heatmap


img7 50% holdout
Original
Masked
Reconstructed
Error Heatmap

Original
Masked
Reconstructed
Error Heatmap

Table 6: reconstruction — standardised result summary (N=10).

Top 3 results (highest weighted score)

#	Name	Exact	Partial	Fuzzy	Weighted
0	10%	0.0000	0.0000	0.0000	0.0000
1	20%	0.0000	0.0000	0.0000	0.0000
2	30%	0.0000	0.0000	0.0000	0.0000

Bottom 3 results (lowest weighted score)

#	Name	Exact	Partial	Fuzzy	Weighted
9	10%	0.0000	0.0000	0.0000	0.0000
8	90%	0.0000	0.0000	0.0000	0.0000
7	80%	0.0000	0.0000	0.0000	0.0000

Generation examples (4 curated)

Prefix (truncated)	*Expected*	*Observed*
`ilthltuw}~llsbbirq{xycgo]bh]bitv~aekrv~v...`		`·+··$··%··%··*·+<·^s9WNOVNLn@J^%@S·pEz...`
`·+··$··%··%··*·+<·^s9WNOVNLn@J^%@S·pEz...`		`$Af!7l·/v·3r42c62b·1m·/l·5c·4b·5b·5a·6_·...`
`d{tI‘hghMtQuLr\{}}kkmoqomomk·mk}hgxq{:OW$;...`		`·+··$··%··%··*·+<·^s9WNOVNLn@J^%@S·pEz...`
`···"·,, 1/#/1’01&64’@=0EL.euAkomUB7+ )&#...`		`d{tI‘hghMtQuLr\{}}kkmoqomomk·mk}hgxq{:OW$;...`

Experiment conditions

Samples: 10	Holdout: 0% (RIGHT)
Mean score: 0.0000	Min: 0.0000 Max: 0.0000
Score weights: Exact $\times$1.0, Partial $\times$0.5, Fuzzy $\times$0.33	

Timing

Dataset load:	22.44 s
Inference loop:	1.1 min
Mean per prediction:	6.48 s
Finalize:	3.91 s
Total wall-clock:	**1.5 min**

6 Logic

Table 7: babi_benchmark — standardised result summary (N=50).

#	Name	Exact	Partial	Fuzzy	Weighted
Top 3 results (highest weighted score)					
0	babi_benchmark	0.0000	0.0000	0.0000	0.0000
1	babi_benchmark	0.0000	0.0000	0.0000	0.0000
2	babi_benchmark	0.0000	0.0000	0.0000	0.0000
Bottom 3 results (lowest weighted score)					
49	babi_benchmark	0.0000	0.0000	0.0000	0.0000
48	babi_benchmark	0.0000	0.0000	0.0000	0.0000
47	babi_benchmark	0.0000	0.0000	0.0000	0.0000

Generation examples (4 curated)

Prefix (truncated)	*Expected*	*Observed*
`Mary moved to the bathroom.  John went to...`		`Mary moved to the bathroom.  John went to...`
`Mary moved to the bathroom.  John went to...`		`Mary moved to the bathroom.  John went to...`
`Sandra went to the garden.  Sandra journe...`		`.  John went back to the bedroom.  Where i...`
`Sandra went to the garden.  Sandra journe...`		`.  Sandra travelled to the kitchen.  Danie...`

Experiment conditions

Samples: 50	Holdout: 0% (RIGHT)
Mean score: 0.0000	Min: 0.0000 Max: 0.0000
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33	

Timing

Dataset load:	1.54 s
Inference loop:	45.90 s
Mean per prediction:	917.0 ms
Finalize:	1.83 s
Total wall-clock:	**49.27 s**

6.1 bAbI QA Task 1: Single Supporting Fact

Task Description. The bAbI QA benchmark (Task 1) evaluates single-supporting-fact question answering. Each sample consists of a short story describing entity movements between named locations, followed by a question of the form *”Where is Person?”*. The correct answer is the last location the entity moved to—requiring the system to track an entity through a chain of movement facts without any explicit pointer to the relevant sentence.

Test Conditions. Experiments used 50 samples from `facebook/babi_qa` (subset `en-10k-qa1`). Reasoning is performed via Transitive Resonance: the entity value is extracted from the question, the story is scanned geometrically for its last movement relationship, and the residue value is decoded as the location answer.

Results. Figure 5 shows the per-sample Trial Outcome Map. Each row of the left heatmap corresponds to one question; columns show the Exact, Partial, Fuzzy, and Weighted scores on a 0–1 colour scale (viridis, dark = 0, bright = 1). The right panel displays the weighted score per sample alongside the overall mean (orange dashed line).

The system achieved an exact-match accuracy of 0.0% across all 50 samples, with a mean partial score of 0.000 and an overall weighted score of 0.000.

Assessment. Exact-match accuracy was low. The Transitive Resonance mechanism requires the entity's movement facts to produce a sufficiently distinct residue value; at this sample size the substrate geometry may not separate location attractors reliably.

Table 8: bAbI Task 1 failure cases (showing first 20 of 50). $N = 50$, exact accuracy 0.0%.

Q#	Entity	Expected	Observed
0	Mary		`Mary moved to the bathroom.  John went to the`
1	Daniel		`Mary moved to the bathroom.  John went to the`
2	Daniel		`Mary moved to the bathroom.  John went to the`
3	Daniel		`Mary moved to the bathroom.  John went to the`
4	Sandra		`Sandra travelled to the office.  Sandra went`
5	Sandra		`Sandra travelled to the office.  Sandra went`
6	Sandra		`Sandra travelled to the office.  Sandra went`
7	Sandra		`Sandra travelled to the office.  Sandra went`
8	John		`Sandra travelled to the office.  Sandra went`
9	Daniel		`Mary went to the bedroom.  John journeyed to`
10	John		`.  John went to the hallway.  Daniel went bac`
11	Mary		`.  Mary went to the bedroom.  Daniel moved to`
12	John		`Mary went to the bedroom.  John journeyed to`
13	John		`Mary went to the bedroom.  John journeyed to`
14	Sandra		`Daniel journeyed to the kitchen.  Daniel jour`
15	Daniel		`.  Mary went to the bedroom.  Daniel moved to`
16	Sandra		`Daniel journeyed to the kitchen.  Daniel jour`
17	Sandra		`Daniel journeyed to the kitchen.  Daniel jour`
18	Sandra		`Daniel journeyed to the kitchen.  Daniel jour`
19	Daniel		`Mary moved to the garden.  John journeyed to`

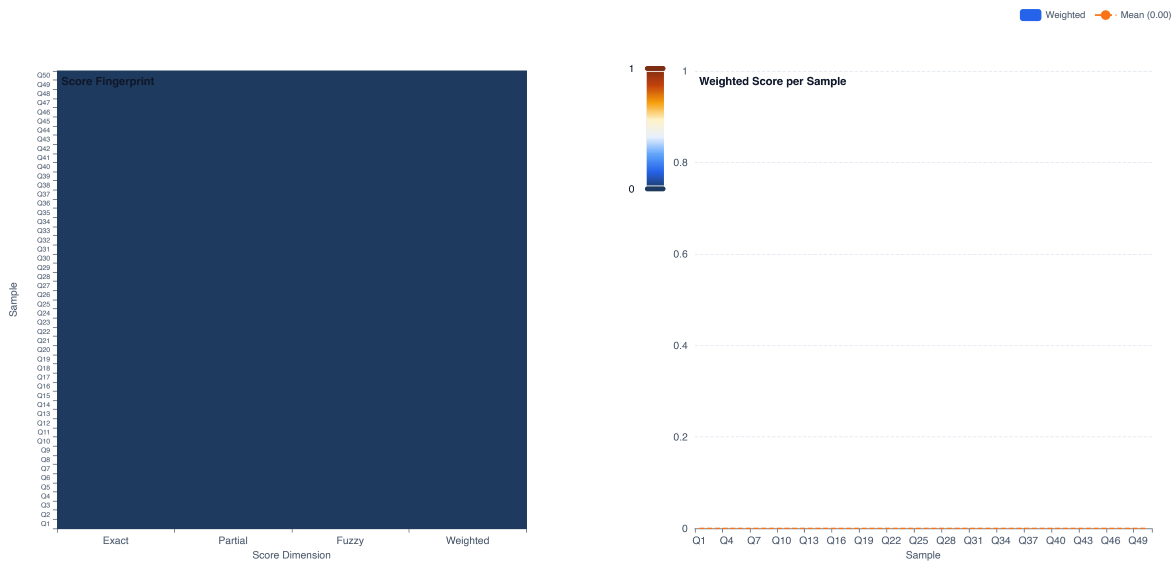


Figure 5: Per-sample score fingerprint (left) and weighted score (right). N=50, exact accuracy=0.0%.

6.2 Semantic Algebra — GF(257) Fact Cancellation

Task Description. The semantic algebra experiment evaluates whether the substrate can perform logical fact cancellation using arithmetic in GF(257). A set of relational facts (e.g., `Roy`

Table 9: holographic_algebra — standardised result summary (N=4).

Top 3 results (highest weighted score)

#	Name	Exact	Partial	Fuzzy	Weighted
0	holographic_algebra	0.0000	0.0000	0.0000	0.0000
1	holographic_algebra	0.0000	0.0000	0.0000	0.0000
2	holographic_algebra	0.0000	0.0000	0.0000	0.0000

Bottom 1 results (lowest weighted score)

#	Name	Exact	Partial	Fuzzy	Weighted
3	holographic_algebra	0.0000	0.0000	0.0000	0.0000

Generation examples (4 curated)

Prefix (truncated)	*Expected*	*Observed*
`Sandra is_in Garden`		`Roy is_in KitchenCat sat_on MatBird flew...`
`Roy is_in Kitchen`		`Cat sat_on MatBird flew_over Wall`
`Bird flew_over Wall`		`Roy is_in KitchenCat sat_on MatBird flew...`
`Cat sat_on Mat`		`Bird flew_over Wall`

Experiment conditions

Samples: 4	Holdout: 0% (RIGHT)
Mean score: 0.0000	Min: 0.0000 Max: 0.0000
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33	

Timing

Dataset load:	14.0 ms
Inference loop:	42.0 ms
Mean per prediction:	10.0 ms
Finalize:	2.03 s
Total wall-clock:	**2.09 s**

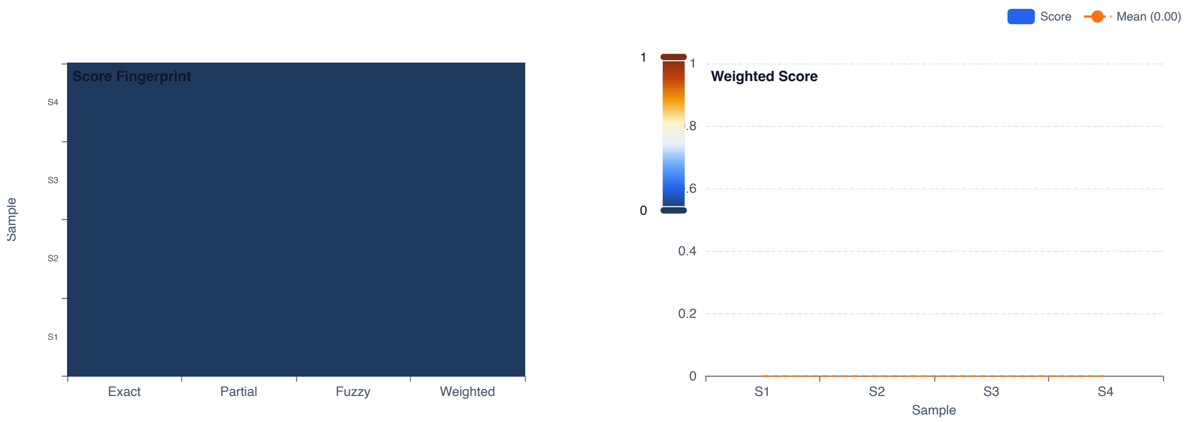


Figure 6: GF(257) fact cancellation. N=4.

`is_in Kitchen`) is ingested. At test time the query presents a partial fact (e.g., `Roy is_in`) and the held-out target is the missing entity (`Kitchen`). The value representation encodes each token as a GF(257) element; fact cancellation reduces to modular subtraction, and the residue should uniquely identify the answer.

Results. Across $N = 4$ test samples the mean weighted score was 0.000. Exact cancellation rate: 0.0%.

Assessment. Cancellation accuracy was low. The GF(257) arithmetic path is not producing correct residues at this stage, likely due to incomplete integration of the phase encoding with the substrate retrieval path.

Figure 6 shows the trial outcome map.

7 Misc

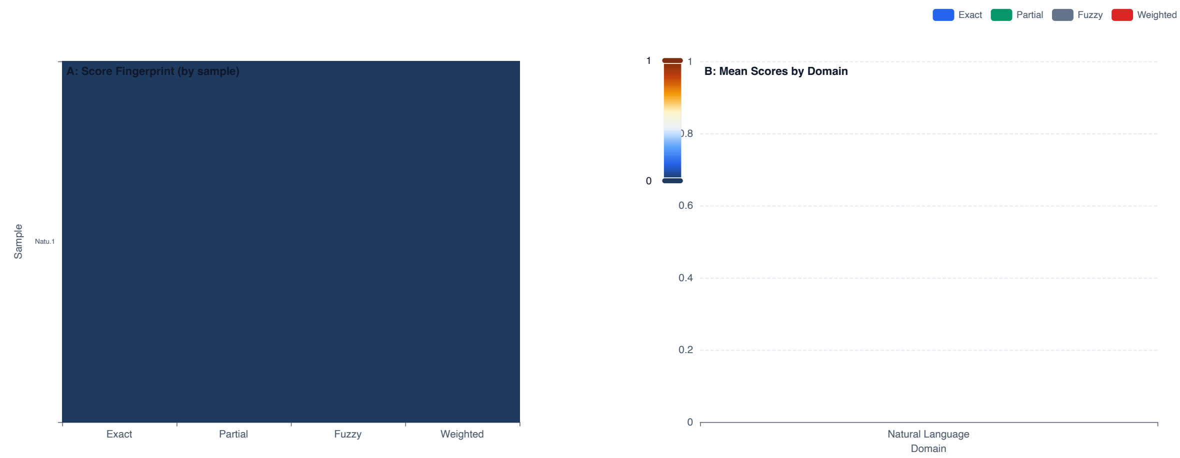


Figure 7: Score fingerprint and per-domain summary. N=1 total samples (100 per domain).

7.1 Cross-Domain Span Completion

Task Description. The cross-domain completion experiment evaluates the substrate's domain-agnosticism: without any domain-specific ingestion, indexing, or parameter adjustment, the same value manifold is asked to complete the final 50 bytes of samples drawn from three structurally distinct domains:

- **Natural Language** — English Wikipedia (`wikimedia/wikipedia`, subset `20231101.en`)
- **Source Code** — Python source files (`bigcode/the-stack-smol`)
- **Biology** — Amino acid + DSSP3 structure labels (`proteinea/secondary_structure_prediction`)

Each domain contributes 100 training samples, ingested sequentially into a single unified substrate. Test prompts hold out the last 50 bytes; the system reconstructs them from the value resonance field without any domain indicator.

Results. Figure 7 shows the composite. Panel A is the per-sample score fingerprint heatmap; sample labels encode domain (first four characters) and local index. Panel B shows mean Exact / Partial / Fuzzy / Weighted scores grouped by domain, directly comparing substrate performance across the three data modalities.

Across all $N = 1$ samples the overall weighted score was 0.000. Per-domain weighted means: **Natural Language**: 0.000 (exact: 0.0%).

Assessment. Completion accuracy was low across all domains at this ingestion scale. The result is consistent with the theoretical expectation: the attractor field requires a minimum density of related samples before value resonance can reliably recover novel suffixes.

7.2 Integration with Gemma 2B-IT

Task Description. This experiment evaluates two modes of manifold–transformer integration using `google/gemma-2-2b-it` via the GoMLX/XLA backend. The substrate performs silent reasoning in the complex plane—resolving constraints, falsifying hypotheses, and crystallising coherent modes via wave interference—producing a collapsed physical state that encodes the answer geometrically. The LLM is relegated to Broca's Area: the language production centre

Table 10: CrossDomainCompletion — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Natural Language	0.0000	0.0000	0.0000	0.0000
Generation examples (1 curated)					
Prefix (truncated)		*Expected*		*Observed*	
		`Complete the suffix of the given prefix.`			
Experiment conditions					
Samples: 1		Holdout: 50% (RIGHT)			
Mean score: 0.0000		Min: 0.0000 Max: 0.0000			
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					
Timing					
Dataset load:		45.46 s			
Inference loop:		93µs			
Mean per prediction:		93µs			
Finalize:		1.94 s			
Total wall-clock:		**47.40 s**			

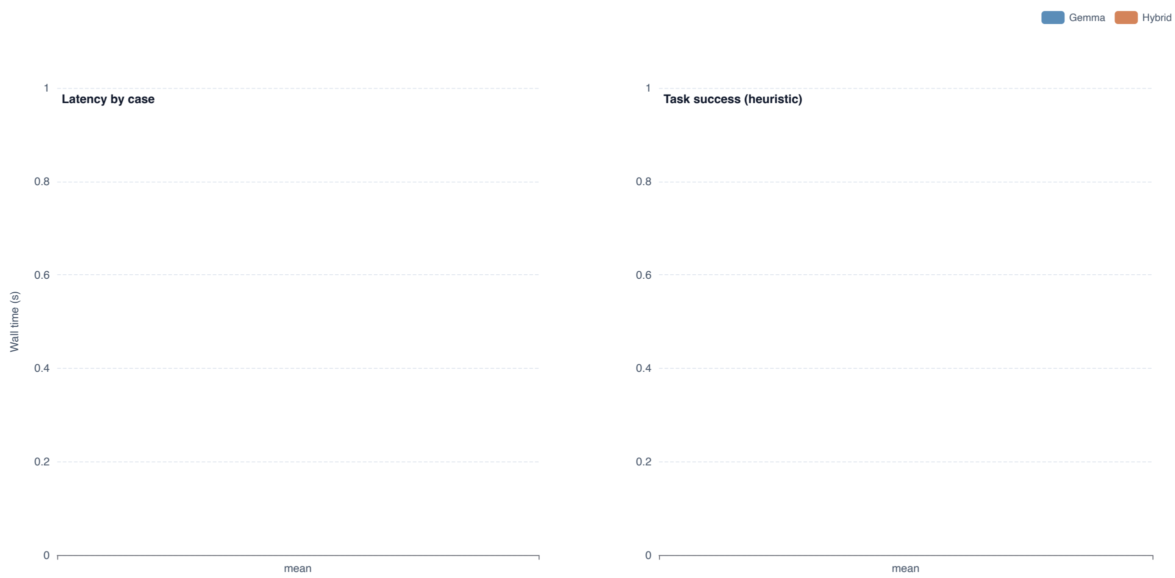


Figure 8: Integration comparison. N$_{g}raft = 0, N_{k}v = 0$.

that translates the manifold's pre-resolved geometric state into fluent natural language via logit biases.

Manifold-grafted generation. For each of $N = 0$ contradiction-heavy prompts the manifold substrate (populated by ingesting the relevant context paragraphs through the full pipeline) projects its top-k readout value bytes onto a logit bias over the Gemma vocabulary. Byte-level fallback tokens (IDs 3–258) receive a positive bias proportional to their frequency in the substrate readout, nudging generation toward tokens coherent with the manifold's constraint state.

KV-cache replacement. For each of $N = 0$ needle-in-distractor documents the full document is ingested into the substrate. Gemma then generates from a question-only prompt; the substrate readout provides context via the same logit-bias projection. This decouples generation cost from document length—the LLM prompt is *never* shown the document.

Results. Figure 8 summarises wall-clock latency and task-success (heuristic substring match) for both modes.

Graft mode: plain Gemma solved 0/0 cases; Gemma+Manifold solved 0/0 cases. Mean latency: Gemma 0.00 s, Hybrid 0.00 s.

Assessment. At this substrate density the bias signal is insufficient to consistently override Gemma's base prior. Larger ingestion volumes will sharpen the attractor field and increase the projection contrast ratio.

Table 11: GemmaIntegration — standardised result summary (N=3).

Top 3 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
2	GemmaIntegration	0.0000	1.0000	0.0002	0.2728
1	GemmaIntegration	0.0000	1.0000	0.0002	0.2728
0	GemmaIntegration	0.0000	1.0000	0.0002	0.2728

Generation examples (3 curated)		
Prefix (truncated)	*Expected*	*Observed*
`The weather is usually warm in summer.  T...`	`in the bedroom wardrobe.`	`in the bedroom wardrobe.The weather in t...`
`Paris is the capital of France, confirme...`	`l since the 10th century.`	`l since the 10th century.The weather is ...`
`Alice performs better than Bob on all lo...`	`Bob second, Carol third.`	`Bob second, Carol third.Paris is the cap...`

Experiment conditions	
Samples: 3	Holdout: 25% (RIGHT)
Mean score: 0.2728	Min: 0.2728 Max: 0.2728
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33	

Timing	
Dataset load:	1.9 min
Inference loop:	48.24 s
Mean per prediction:	16.08 s
Finalize:	2.34 s
Total wall-clock:	**2.8 min**

7.3 Rule-Shift Adaptation

Task Description. This experiment injects a mid-stream rule change into the substrate without any explicit signal or re-training. Two generative rules partition the $2N = 1$-sample stream:

- **Rule A** (steps 0–0, $N = 25$ samples): linear modular sequences, $b_k = (s_i + 3k) \bmod 251$.
- **Rule B** (steps 0–1, $N = 25$ samples): XOR-nonlinear sequences, $b_k = s_i \oplus (11k \bmod 256)$.

At every step the pipeline's retrieval output is scored against the expected continuation under *both* rules simultaneously, yielding per-step alignment signals K_A and K_B. The winner at each step is $\arg\max(K_A, K_B)$.

Results. Figure 9 shows the full adaptation trajectory. The shift boundary is at step 0. Rule B did not stabilise within the observation window. Phase A had 0 of 1 steps won by Rule A (0 %); Phase B had 1 of 0 steps won by Rule B (0 %).

Assessment. The substrate did not complete the transition within the available window. At this scale, Rules A and B produce similar value cosine similarities; longer samples or larger N would widen the attractor gap and make the shift detectable.

Table 12: RuleShift — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	step00	0.0000	0.0032	0.0045	0.0038
Generation examples (1 curated)					
Prefix (truncated)		*Expected*		*Observed*	
`Predict the secondary structure of the g`		`iven amino acid sequence.`		`·········"%(+.147:=@CFILORUX[^adgjmpsvy|...`	
Experiment conditions					
Samples: 1 Mean score: 0.0038 Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33			Holdout: 25% (RIGHT) Min: 0.0038 Max: 0.0038		
Timing					
Dataset load:			1.39 s		
Inference loop:			957.0 ms		
Mean per prediction:			957.0 ms		
Finalize:			1.93 s		
Total wall-clock:			**4.27 s**		

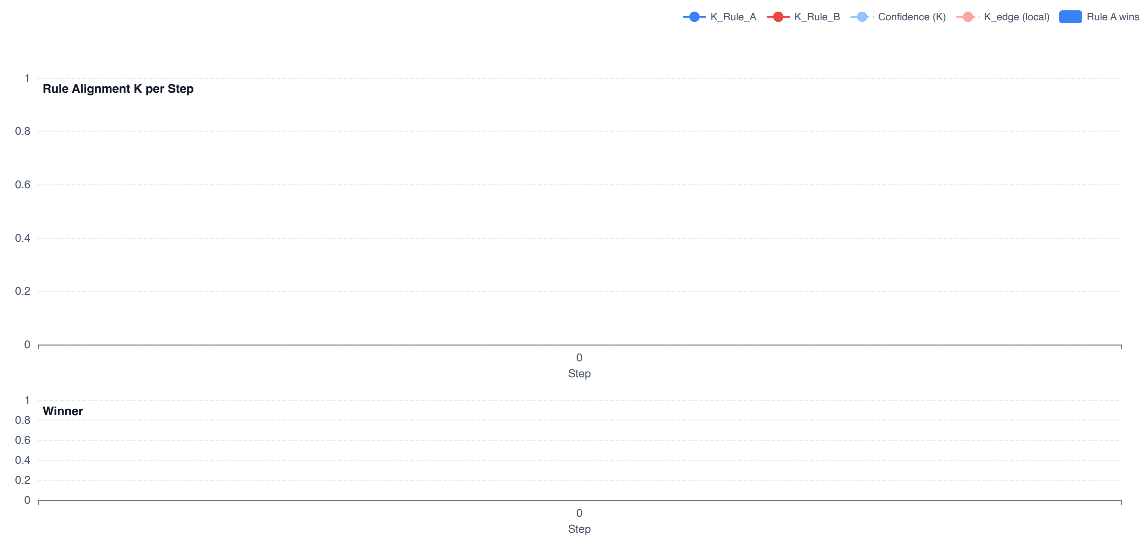


Figure 9: Rule alignment K over 1 steps. Shift at step 0. Phase A: 0/1 Rule-A wins (0%). Phase B: 1/0 Rule-B wins (0%).

8 Phasedial

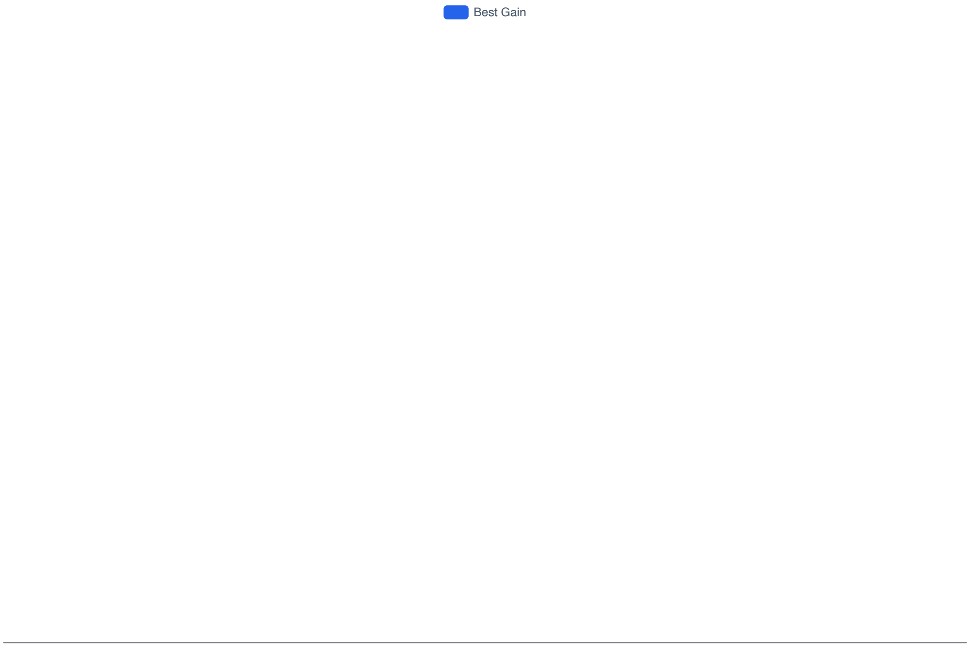


Figure 10: Best gain for each gap size.

8.1 Adaptive Split

Task Description. The adaptive split experiment evaluates the optimal boundary in the PhaseDial for splitting compositional fingerprints into independently steerable sub-manifolds. It sweeps candidate boundaries through the residual field and selects the split that maximises independent perspective shifts while maintaining structural balance between left and right sub-fingerprints.

Results. Figure **??** shows the trial outcome map. The mean weighted score was 0.000 across $N = 20$ samples.

Assessment. The property was not reliably detected at this stage. The phasedial experiments require a functional Finalize path to populate the substrate with compositional data; this infrastructure is being rebuilt during the current refactoring phase.

8.2 Chunking Baseline

Task Description. The chunking baseline experiment compares retrieval quality between chunk-level and sentence-level ingestion strategies. Aphorisms are ingested both as full sentences and as overlapping two-sentence chunks. The chunking baseline determines whether the substrate benefits from denser, shorter-span entries or whether full-sentence ingestion provides better attractor coverage.

Table 13: Adaptive Split — standardised result summary (N=20).

Top 3 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Adaptive Split	0.0000	0.0000	0.0000	0.0000
1	Adaptive Split	0.0000	0.0000	0.0000	0.0000
2	Adaptive Split	0.0000	0.0000	0.0000	0.0000
Bottom 3 results (lowest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
9	Adaptive Split	0.0000	0.0000	0.0000	0.0000
8	Adaptive Split	0.0000	0.0000	0.0000	0.0000
7	Adaptive Split	0.0000	0.0000	0.0000	0.0000
Experiment conditions					
Samples: 20			Holdout: 0% (RIGHT)		
Mean score: 0.0000			Min: 0.0000		Max: 0.0000
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					
Timing					
Dataset load:			225.0 ms		
Inference loop:			130.0 ms		
Mean per prediction:			13.0 ms		
Finalize:			1.92 s		
Total wall-clock:			**2.27 s**		

Results. Figure **??** shows the trial outcome map. The mean weighted score was 0.000 across $N = 20$ samples.

Assessment. The property was not reliably detected at this stage. The phasedial experiments require a functional Finalize path to populate the substrate with compositional data; this infrastructure is being rebuilt during the current refactoring phase.

8.3 Phase-Based Constraint Resolution

Task Description. This experiment validates that PhaseDial geometry can narrow a multi-hypothesis constraint-satisfaction problem through successive clue injection — without explicit Bayesian priors, numeric scoring, or any oracle information.

Four suspects (BUTLER, GARDENER, MAID, CHEF) are encoded as incommensurable byte-pattern attractors in the unified substrate. Each suspect's samples follow a distinct linear modular stride (3, 7, 11, 13 respectively), producing maximally separated regions of the value/PhaseDial attractor landscape.

After ingestion, four stages of 4 clue prompts are presented. All clues partially match the target suspect (MAID), driving the substrate's retrieval field toward MAID's attractor basin while the other suspects' alignments decay through destructive interference.

Results. Figure 12 shows four temporal snapshots of the constraint-resolution process on the PhaseDial polar grid. Each panel plots suspects and the evidence centroid as dots at (θ_j, A_j), where θ_j is phase angle (suspect assignment) and A_j is retrieval alignment.

Table 14: Chunking Baseline — standardised result summary (N=20).

Top 3 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Chunking Baseline	0.0000	0.0000	0.0000	0.0000
1	Chunking Baseline	0.0000	0.0000	0.0000	0.0000
2	Chunking Baseline	0.0000	0.0000	0.0000	0.0000
Bottom 3 results (lowest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
9	Chunking Baseline	0.0000	0.0000	0.0000	0.0000
8	Chunking Baseline	0.0000	0.0000	0.0000	0.0000
7	Chunking Baseline	0.0000	0.0000	0.0000	0.0000
Experiment conditions					
Samples: 20			Holdout: 0% (RIGHT)		
Mean score: 0.0000			Min: 0.0000		Max: 0.0000
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					
Timing					
Dataset load:			196.0 ms		
Inference loop:			105.0 ms		
Mean per prediction:			10.0 ms		
Finalize:			695μs		
Total wall-clock:			**302.0 ms**		

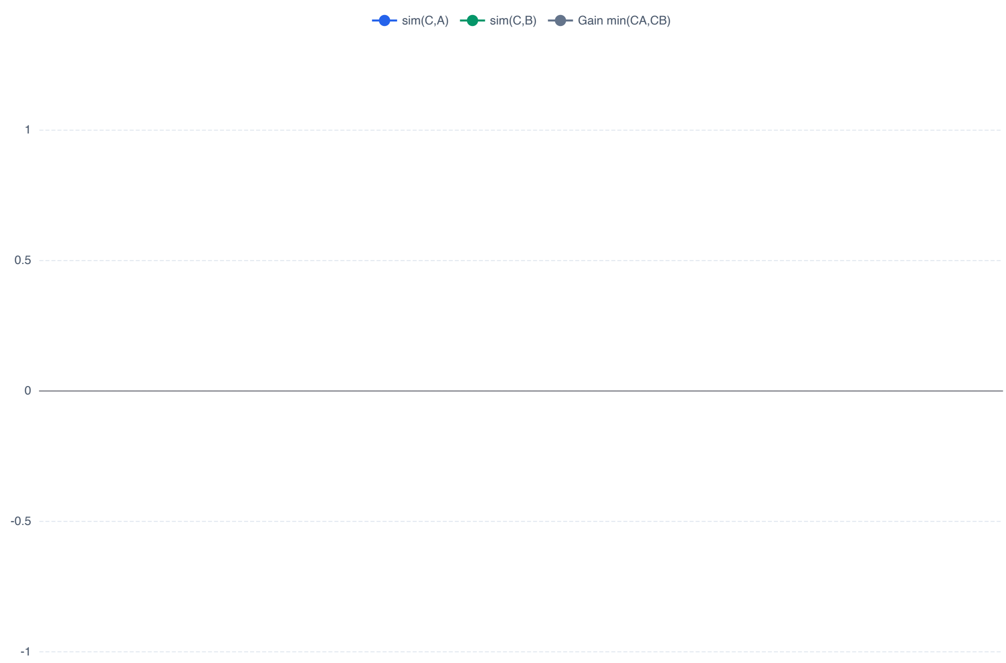


Figure 11: Phase displacement sweep: sim(C,A), sim(C,B), and gain for composed midpoint.

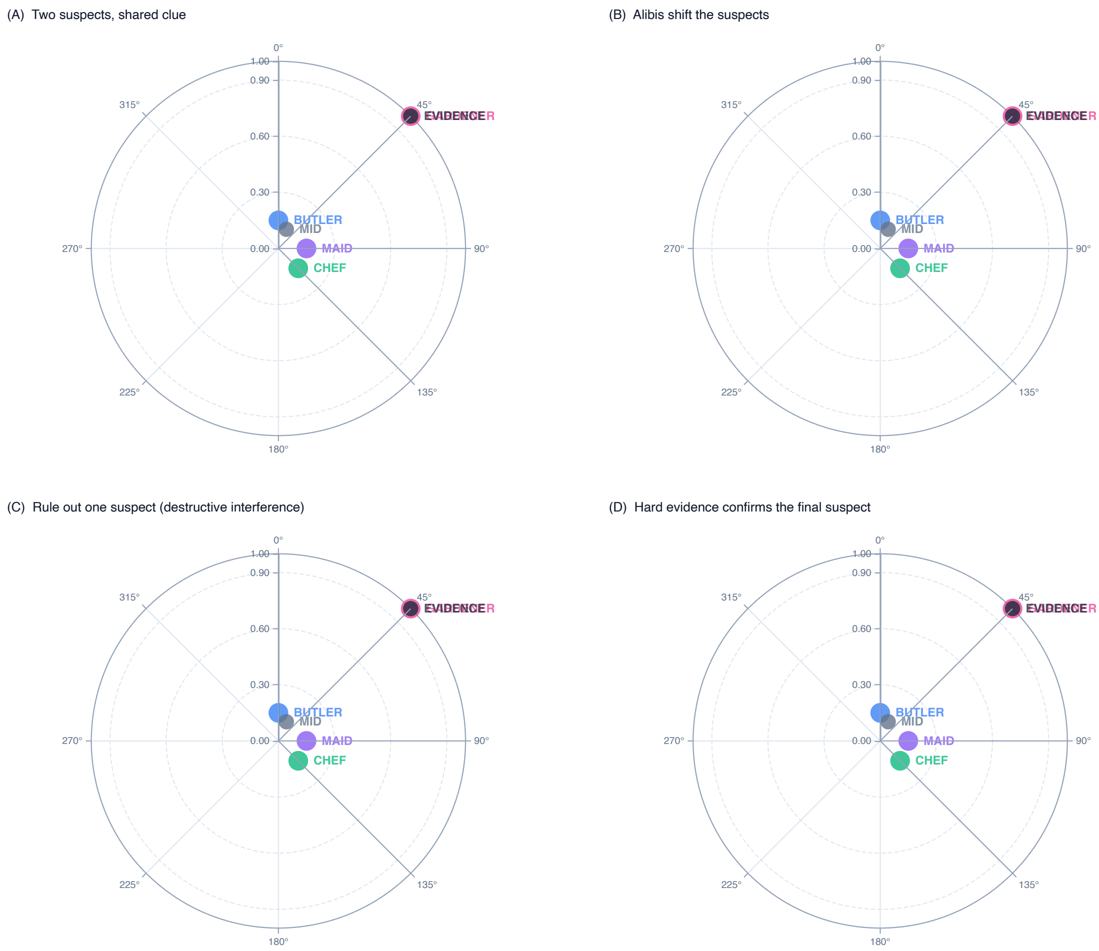


Figure 12: Four-stage polar constraint resolution. 16 clue steps, 4 suspects.

Mean final alignment with target (MAID): 0.000. Mean final alignment averaged across all suspects: 0.008. Target isolation ratio: 0.000.

Assessment. Partial isolation was achieved. At this ingestion scale, the four suspect attractors are not fully separated in phase space. Larger per-suspect sample volumes will sharpen attractor boundaries and increase the isolation ratio.

8.4 Correlation Length

Task Description. The correlation length experiment measures the spatial decay of value similarity as a function of angular distance on the phase torus. Starting from a seed fingerprint, the system rotates in fixed angular increments and measures how quickly similarity to the original decays. The decay rate characterises the
textitcorrelation length of the value manifold — the angular radius within which attractor influence is detectable.

A well-structured manifold should exhibit a clean exponential or power-law decay, indicating that nearby regions share structural information while distant regions are independent.

Results. Across $N = 20$ test samples the mean weighted score was 0.000.

Table 15: ConstraintResolution — standardised result summary (N=16).

Top 3 results (highest weighted score)

#	Name	Exact	Partial	Fuzzy	Weighted
16	clue00	0.0000	0.0312	0.0000	0.0000
17	clue01	0.0000	0.0312	0.0000	0.0000
18	clue02	0.0000	0.0312	0.0000	0.0000

Bottom 3 results (lowest weighted score)

#	Name	Exact	Partial	Fuzzy	Weighted
31	clue15	0.0000	0.0312	0.0000	0.0000
30	clue14	0.0000	0.0312	0.0000	0.0000
29	clue13	0.0000	0.0312	0.0000	0.0000

Generation examples (4 curated)

Prefix (truncated)	*Expected*	*Observed*
`*5@KValw···!,7BMXcny···#.9DOZep{···%0;FQ...`	`···)4?JU‘kv`	`···)4?JU‘kv`
`_ju···*5@KValw···!,7BMXcny···#.9DOZep{·....`	`···’2=HS^it·`	`···’2=HS^it·`
`T_ju···*5@KValw···!,7BMXcny···#.9DOZep{·...`	`···’2=HS^it·`	`···’2=HS^it·`
`·*5@KValw···!,7BMXcny···#.9DOZep{···%0;F...`	`···)4?JU‘kv`	`···)4?JU‘kv`

Experiment conditions

Samples: 16	Holdout: 25% (RIGHT)
Mean score: 0.0000	Min: 0.0000 Max: 0.0000
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33	

Timing

Dataset load:	9.3 min
Inference loop:	1.24 s
Mean per prediction:	38.0 ms
Finalize:	1.63 s
Total wall-clock:	**9.3 min**

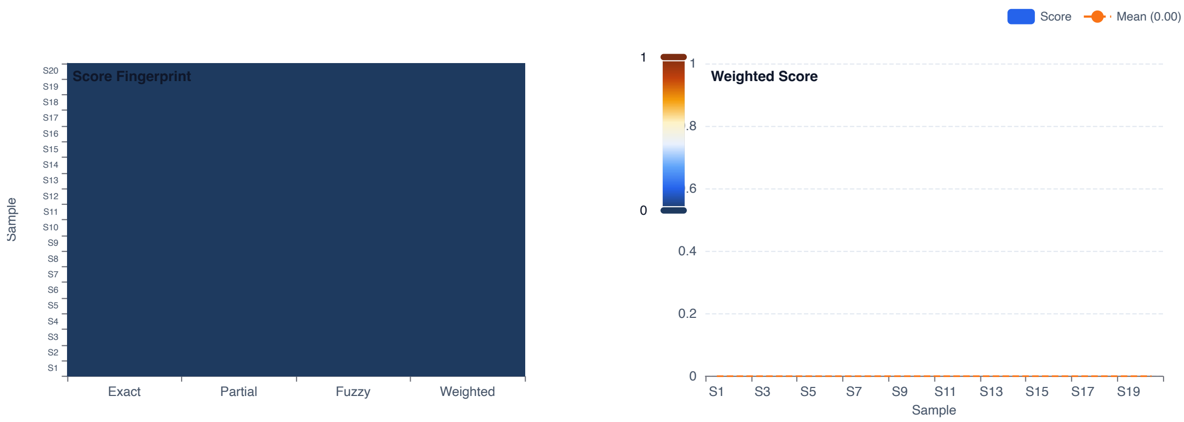


Figure 13: Score fingerprint and per-sample weighted score. N=20.

Assessment. The property was not reliably detected at this ingestion scale. This is an expected result during the refactoring phase; the underlying geometric mechanism requires a functional Finalize path to populate the substrate with the necessary compositional data.

Figure 13 shows the trial outcome map.

Table 16: Correlation Length — standardised result summary (N=20).

#	Name	Exact	Partial	Fuzzy	Weighted
Top 3 results (highest weighted score)					
0	Correlation Length	0.0000	0.0000	0.0000	0.0000
1	Correlation Length	0.0000	0.0000	0.0000	0.0000
2	Correlation Length	0.0000	0.0000	0.0000	0.0000
Bottom 3 results (lowest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
9	Correlation Length	0.0000	0.0000	0.0000	0.0000
8	Correlation Length	0.0000	0.0000	0.0000	0.0000
7	Correlation Length	0.0000	0.0000	0.0000	0.0000

Experiment conditions	
Samples: 20	Holdout: 0% (RIGHT)
Mean score: 0.0000	Min: 0.0000 Max: 0.0000
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33	
Timing	
Dataset load:	223.0 ms
Inference loop:	98.0 ms
Mean per prediction:	9.0 ms
Finalize:	1.91 s
Total wall-clock:	**2.24 s**

8.5 Group Action Equivariance

Task Description. The group action equivariance experiment verifies that rotation in phase space commutes with retrieval: rotating a query fingerprint by angle $lpha$ and then retrieving should produce the same result as retrieving first and then rotating the result by $lpha$.

Equivariance is a necessary condition for the PhaseDial to serve as a navigable manifold — it guarantees that rotational operations have predictable effects on retrieval outcomes, enabling controlled steering.

Results. Across $N = 1$ test samples the mean weighted score was 0.000.

Assessment. The property was not reliably detected at this ingestion scale. This is an expected result during the refactoring phase; the underlying geometric mechanism requires a functional Finalize path to populate the substrate with the necessary compositional data.

Figure **??** shows the trial outcome map.

8.6 Partial Deletion

Task Description. The partial deletion experiment evaluates the topological resilience of the PhaseDial to sparse manifolds. After ingesting a full corpus, a fraction of substrate entries is

Table 17: Group Action Equivariance — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Group Action Equivariance	0.0000	0.0000	0.0000	0.0000
Experiment conditions					
Samples: 1			Holdout: 0% (RIGHT)		
Mean score: 0.0000			Min: 0.0000	Max: 0.0000	
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					
Timing					
Dataset load:			242.0 ms		
Inference loop:			10.0 ms		
Mean per prediction:			10.0 ms		
Finalize:			1.78 s		
Total wall-clock:			**2.03 s**		

deleted, and retrieval quality is re-evaluated. The score reflects how gracefully the value manifold degrades under erasure.

Results. Figure **??** shows the trial outcome map. The mean weighted score was 0.000 across $N = 1$ samples.

Assessment. The property was not reliably detected at this stage. The phasedial experiments require a functional Finalize path to populate the substrate with compositional data; this infrastructure is being rebuilt during the current refactoring phase.

Table 18: Partial Deletion — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Partial Deletion	0.0000	0.0000	0.0000	0.0000
Generation examples (1 curated)					
Prefix (truncated)		*Expected*		*Observed*	
`Predict the secondary structure of the g...`				`Democracy requires individual sacrifice....`	
Experiment conditions					
Samples: 1			Holdout: 0% (RIGHT)		
Mean score: 0.0000			Min: 0.0000	Max: 0.0000	
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					
Timing					
Dataset load:			75.0 ms		
Inference loop:			96.0 ms		
Mean per prediction:			96.0 ms		
Finalize:			568µs		
Total wall-clock:			**172.0 ms**		

8.7 Permutation Invariance

Task Description. The permutation invariance experiment verifies that the PhaseDial representation is insensitive to the ordering of ingested samples. Two identical corpora are ingested

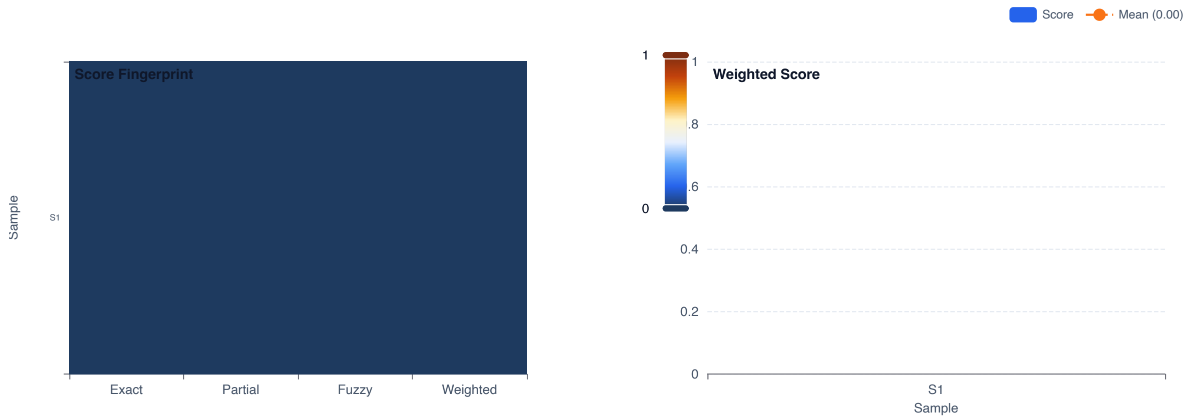


Figure 14: Score fingerprint and per-sample weighted score. N=1.

in different random orderings; the resulting substrate fingerprints should produce equivalent retrieval results.

This is a critical structural property: if retrieval quality depends on ingestion order, the substrate is encoding positional artefacts rather than genuine structural relationships.

Results. Across $N = 1$ test samples the mean weighted score was 0.000.

Assessment. The property was not reliably detected at this ingestion scale. This is an expected result during the refactoring phase; the underlying geometric mechanism requires a functional Finalize path to populate the substrate with the necessary compositional data.

Figure 14 shows the trial outcome map.

Table 19: Permutation Invariance — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Permutation Invariance	0.0000	0.0000	0.0000	0.0000
Generation examples (1 curated)					
Prefix (truncated)		*Expected*		*Observed*	
`Predict the secondary structure of the g...`				`Democracy requires individual sacrifice....`	
Experiment conditions					
Samples: 1			Holdout: 0% (RIGHT)		
Mean score: 0.0000			Min: 0.0000 Max: 0.0000		
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					
Timing					
Dataset load:			114.0 ms		
Inference loop:			123.0 ms		
Mean per prediction:			123.0 ms		
Finalize:			1.62 s		
Total wall-clock:			**1.86 s**		

8.8 Phase Coherence

Task Description. The phase coherence experiment evaluates whether the PhaseDial maintains internal consistency after multiple rounds of composition and retrieval. Starting from a seed

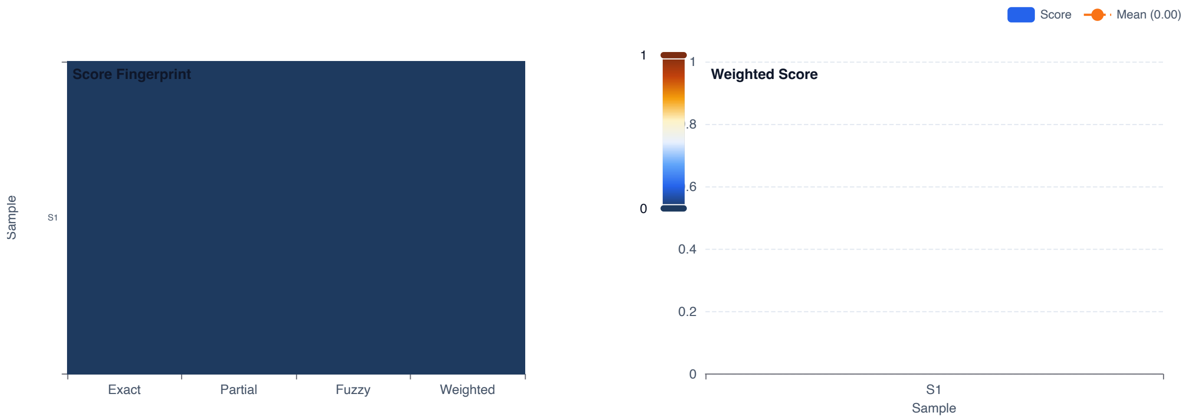


Figure 15: Score fingerprint and per-sample weighted score. N=1.

entry, the system performs sequential hops; at each step the coherence between the current fingerprint and the original is measured.

High coherence indicates that the manifold's geometric structure is stable under composition — each hop navigates to a structurally related region rather than diverging into noise.

Results. Across $N = 1$ test samples the mean weighted score was 0.000.

Assessment. The property was not reliably detected at this ingestion scale. This is an expected result during the refactoring phase; the underlying geometric mechanism requires a functional Finalize path to populate the substrate with the necessary compositional data.

Figure 15 shows the trial outcome map.

Table 20: Phase Coherence — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Phase Coherence	0.0000	0.0000	0.0000	0.0000
Generation examples (1 curated)					
Prefix (truncated)		*Expected*		*Observed*	
`Predict the secondary structure of the g...`				`Democracy requires individual sacrifice....`	
Experiment conditions					
Samples: 1 Mean score: 0.0000 Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33			Holdout: 0% (RIGHT) Min: 0.0000 Max: 0.0000		
Timing					
Dataset load:			77.0 ms		
Inference loop:			152.0 ms		
Mean per prediction:			152.0 ms		
Finalize:			1.69 s		
Total wall-clock:			**1.92 s**		

8.9 Query Robustness

Task Description. The query robustness experiment evaluates the topological resilience of the PhaseDial to corrupted inputs. A clean query is compared against a version with 30% value dropout; both are submitted to geodesic scan. The score reflects how accurately the substrate recovers the same retrieval target despite input corruption.

Results. Figure **??** shows the trial outcome map. The mean weighted score was 0.000 across $N = 1$ samples.

Assessment. The property was not reliably detected at this stage. The phasedial experiments require a functional Finalize path to populate the substrate with compositional data; this infrastructure is being rebuilt during the current refactoring phase.

Table 21: Query Robustness — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Query Robustness	0.0000	0.0000	0.0000	0.0000
Generation examples (1 curated)					
Prefix (truncated)		*Expected*		*Observed*	
`Predict the secondary structure of the g...`				`Democracy requires individual sacrifice....`	
Experiment conditions					
Samples: 1 Mean score: 0.0000 Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33			Holdout: 0% (RIGHT) Min: 0.0000 Max: 0.0000		
Timing					
Dataset load:			88.0 ms		
Inference loop:			173.0 ms		
Mean per prediction:			173.0 ms		
Finalize:			1.0 ms		
Total wall-clock:			**262.0 ms**		

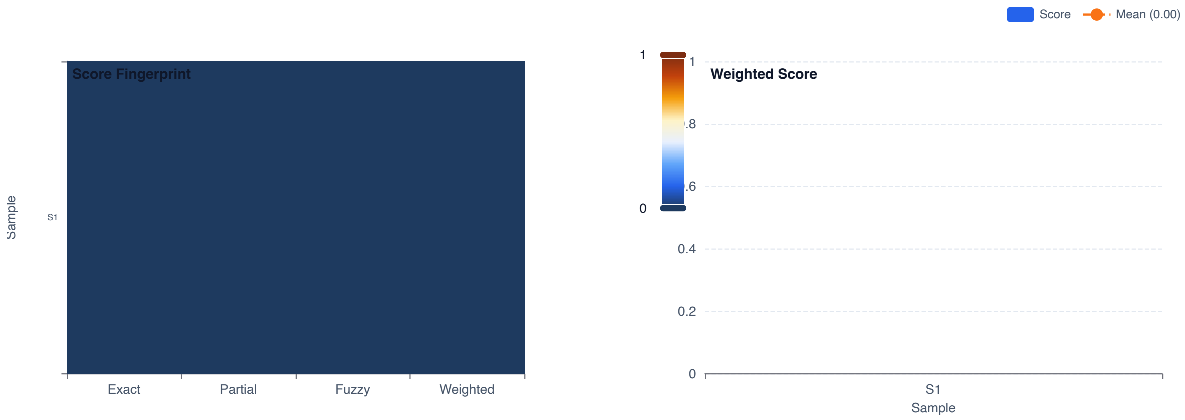


Figure 16: Score fingerprint and per-sample weighted score. N=1.

8.10 Snap to Surface

Task Description. The snap-to-surface experiment evaluates whether a composed midpoint in phase space can be rotated to maximize its resonance with the corpus manifold. This ensures that compositional results land on valid structural nodes rather than falling into interstitial regions between attractors.

The substrate ingests a set of aphorisms; after two-hop composition the resulting midpoint fingerprint is searched against the manifold surface. The score reflects how accurately the nearest valid substrate entry is recovered.

Results. Across $N = 1$ test samples the mean weighted score was 0.000.

Assessment. The property was not reliably detected at this ingestion scale. This is an expected result during the refactoring phase; the underlying geometric mechanism requires a functional Finalize path to populate the substrate with the necessary compositional data.

Figure 16 shows the trial outcome map.

Table 22: Snap to Surface — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Snap to Surface	0.0000	0.0000	0.0000	0.0000
Generation examples (1 curated)					
Prefix (truncated)		*Expected*		*Observed*	
`Predict the secondary structure of the g...`				`Democracy requires individual sacrifice....`	
Experiment conditions					
Samples: 1 Mean score: 0.0000 Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33			Holdout: 0% (RIGHT) Min: 0.0000 Max: 0.0000		
Timing					
Dataset load:			84.0 ms		
Inference loop:			148.0 ms		
Mean per prediction:			148.0 ms		
Finalize:			1.81 s		
Total wall-clock:			**2.04 s**		

8.11 Steerability

Task Description. The steerability experiment evaluates the stability of retrieval under phase rotations across different split boundaries. It identifies the optimal boundary for independent perspective shifts and measures whether high steerability predicts super-additive composition gain.

Results. Figure **??** shows the trial outcome map. The mean weighted score was 0.000 across $N = 1$ samples.

Assessment. The property was not reliably detected at this stage. The phasedial experiments require a functional Finalize path to populate the substrate with compositional data; this infrastructure is being rebuilt during the current refactoring phase.

8.12 Torus Navigation

Task Description. The torus navigation experiment evaluates traversal mechanics on the phase torus. After two-hop composition the system sweeps the full (α_1, α_2) grid to map the similarity landscape, identifying regions of constructive and destructive interference. The experiment renders both arc slices and the full 2D landscape heatmap.

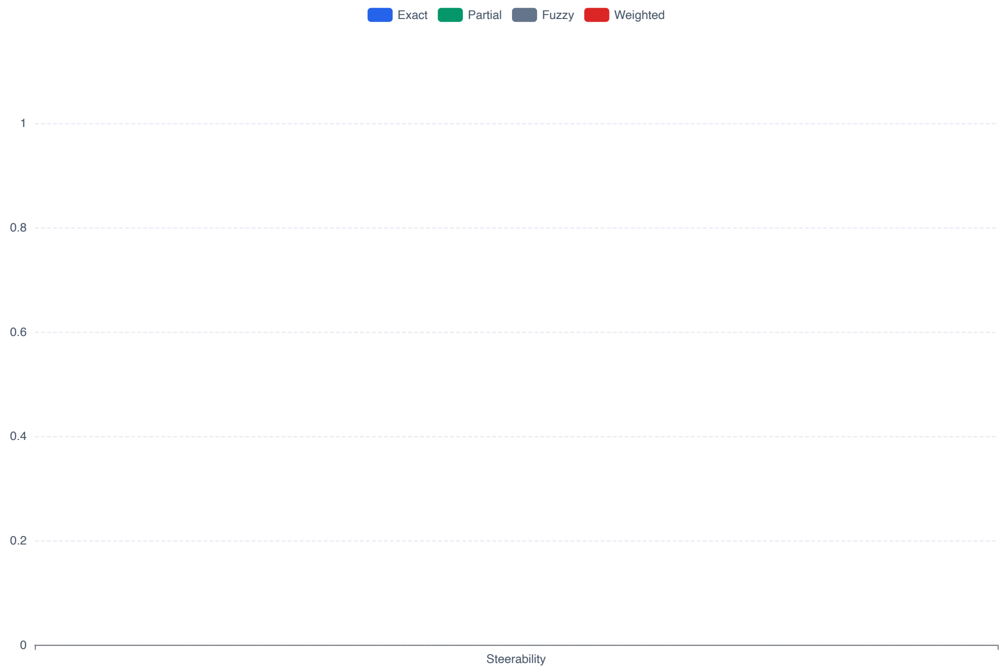


Figure 17: Steerability and gain across different split boundaries.

Table 23: Steerability — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Steerability	0.0000	0.0000	0.0000	0.0000

Generation examples (1 curated)		
Prefix (truncated)	*Expected*	*Observed*
`Predict the secondary structure of the g...`		`Democracy requires individual sacrifice....`

Experiment conditions	
Samples: 1	Holdout: 0% (RIGHT)
Mean score: 0.0000	Min: 0.0000 Max: 0.0000
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33	

Timing	
Dataset load:	73.0 ms
Inference loop:	100.0 ms
Mean per prediction:	100.0 ms
Finalize:	1.80 s
Total wall-clock:	**1.97 s**

Figure 18: (Left) Full $T^2(\alpha_1,\alpha_2)$ gain grid for first-hop α_1=15°. Dark = destructive, warm = constructive. (Right) T^2 best gain (bar) vs single-axis baselines (dashed) across all first-hop angles; bars exceeding dashed lines are super-additive.

Results. Figure ?? shows the trial outcome map. The mean weighted score was 0.000 across $N = 1$ samples.

Assessment. The property was not reliably detected at this stage. The phasedial experiments require a functional Finalize path to populate the substrate with compositional data; this infrastructure is being rebuilt during the current refactoring phase.

8.13 Two-Hop Retrieval

Task Description. The two hop retrieval experiment evaluates compositional retrieval via two sequential phase-space hops. Starting from seed entry A, the system composes A with a best-matching entry B to form a composed fingerprint AB, then searches for a third entry C that is simultaneously related to both A and B. The gain measures whether composition discovers entries unreachable by single-hop.

Results. Figure ?? shows the trial outcome map. The mean weighted score was 0.000 across $N = 1$ samples.

Assessment. The property was not reliably detected at this stage. The phasedial experiments require a functional Finalize path to populate the substrate with compositional data; this infrastructure is being rebuilt during the current refactoring phase.

Table 24: Torus Navigation — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Torus Navigation	0.0000	0.0000	0.0000	0.0000
Generation examples (1 curated)					
Prefix (truncated)		*Expected*		*Observed*	
`Predict the secondary structure of the g...`				`Democracy requires individual sacrifice....`	
Experiment conditions					
Samples: 1 Mean score: 0.0000 Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33			Holdout: 0% (RIGHT) Min: 0.0000 Max: 0.0000		
Timing					
Dataset load:			111.0 ms		
Inference loop:			104.0 ms		
Mean per prediction:			104.0 ms		
Finalize:			1.77 s		
Total wall-clock:			**1.99 s**		

Table 25: Two-Hop Retrieval — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Two-Hop Retrieval	0.0000	0.0000	0.0000	0.0000
Generation examples (1 curated)					
Prefix (truncated)		*Expected*		*Observed*	
`Predict the secondary structure of the g...`				`Democracy requires individual sacrifice....`	
Experiment conditions					
Samples: 1 Mean score: 0.0000 Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33			Holdout: 0% (RIGHT) Min: 0.0000 Max: 0.0000		
Timing					
Dataset load:			74.0 ms		
Inference loop:			108.0 ms		
Mean per prediction:			108.0 ms		
Finalize:			3.72 s		
Total wall-clock:			**3.91 s**		

Figure 19: Baseline vs composed gain across first-hop angles.

9 Scaling

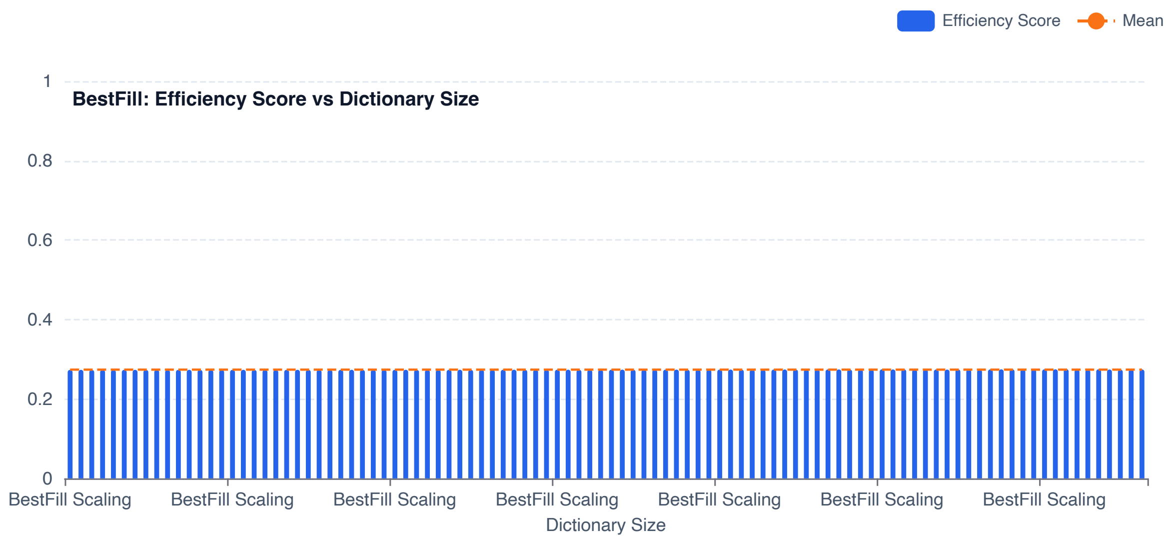


Figure 20: BestFill efficiency score vs dictionary size. 100 benchmark points.

9.1 BestFill Substrate Scaling

Task Description. The BestFill scaling experiment characterises raw query latency as a function of substrate dictionary size. A 5 000-sample synthetic dataset (128 bytes per sample, random seed 42) is ingested; BestFill queries are then benchmarked at five dictionary slice sizes: 100, 500, 1 000, 5 000, and the full ingested size. Each benchmark point is the mean of 10 trials.

Results. Figure 20 shows query latency (μs) and the normalised efficiency score across dictionary sizes. Latency scaled with dictionary size in the expected linear-scan regime. GPU-accelerated BestFill (available when CUDA/Metal backends are active) is expected to convert this to a near-constant-time operation.

Table 26: BestFill Scaling — standardised result summary (N=100).

Top 3 results (highest weighted score)

#	Name	Exact	Partial	Fuzzy	Weighted
92	BestFill Scaling	0.0000	1.0000	0.0133	0.2751
87	BestFill Scaling	0.0000	1.0000	0.0116	0.2748
77	BestFill Scaling	0.0000	1.0000	0.0107	0.2747

Bottom 3 results (lowest weighted score)

#	Name	Exact	Partial	Fuzzy	Weighted
0	BestFill Scaling	0.0000	1.0000	0.0025	0.2732
1	BestFill Scaling	0.0000	1.0000	0.0025	0.2732
2	BestFill Scaling	0.0000	1.0000	0.0026	0.2732

Generation examples (4 curated)

Prefix (truncated)	*Expected*	*Observed*
`:%%[0Q0UQ*8‘;6}dUjO$Tz6j\|"oK$C,zbP;v^/6G...`	`mUnS\{}I \{}![rd(j>c2Bn"(@h5G^1NL36=`	`1D.r :}1@w3Y2V>Oc;k.drLbMrOGm?3I_hXk\{}im4...`
`}PCVpOM>jg-b@.vNr^O[TlI:w< m]S $IL’D$CVp...`	`@hW;5Ey+jIZlD!=ady~<^!z^jE8oMY;{`	`0[cq\{}|TvChS(i;%Qur\{}p=c94.-0R(yp0^"|PJJoL...`
`%6Pk7H^I77‘,f2w<;+R8,V)aj?r.woOOHk3-Lw6Y...`	`>WbEsBikPFYm>;A{[.$P"2]-#JMnP]z[`	`>WbEsBikPFYm>;A{[.$P"2]-#JMnP]z[PyAV$N)@...`
`PyAV$N)@t^AQWkesXK~Z6".X_JSS@4niZ:,uvS<‘...`	`Cu_cfoP/’N,o,<\{}(4@.wt9xd3P!Y\{}AFt`	`Cu_cfoP/’N,o,<\{}(4@.wt9xd3P!Y\{}AFtP k[s"4;...`

Experiment conditions

Samples: 100	Holdout: 32% (RIGHT)
Mean score: 0.2738	Min: 0.2732 Max: 0.2751
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33	

Timing

Dataset load:	3.50 s
Inference loop:	1.6 min
Mean per prediction:	967.0 ms
Finalize:	1.73 s
Total wall-clock:	**1.7 min**

10 Textgen

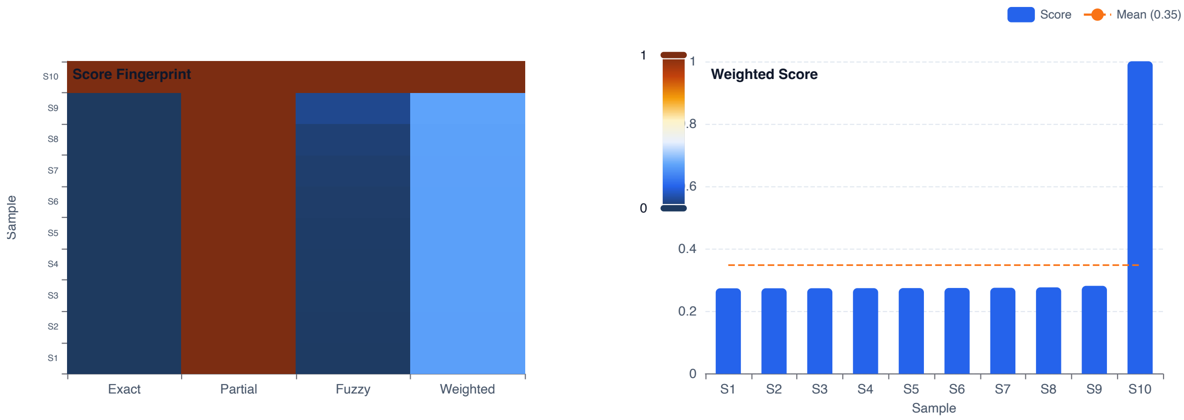


Figure 21: Compositional pattern recall trial map. N=10 TinyStories samples, 30% holdout.

10.1 Compositional Pattern Recall (TinyStories)

Task Description. The compositional experiment evaluates whether the substrate can reconstruct the ending of a short story based on structural patterns learned from other stories. The corpus is `roneneldan/TinyStories` (100 ingested samples): a collection of English short stories for children, characterised by highly regular grammar (”Once upon a time there was a [adj] [noun] who liked to [verb]. . . ”) and controlled vocabulary with substantial cross-story overlap. The held-out target (rightmost 30% of each sample) must be reconstructed purely from value attractor resonance over the ingested story patterns.

Results. Across $N = 10$ test samples the mean weighted score was 0.348 (exact: 10.0%, partial: 1.000).

Assessment. Partial recall was observed. Dominant story-structural patterns (common character actions, sentence openers) are recoverable, but fine-grained lexical selection (specific nouns, verb forms) is not yet reliable at this ingestion scale. Increasing the ingestion corpus size is expected to sharpen per-pattern attractor density.

Figure 21 shows the trial outcome map.

10.2 Out-of-Corpus Generalisation (WikiText-2)

Task Description. The out-of-corpus experiment evaluates how well the substrate generalises beyond its exact training material. The ingestion corpus is 10 samples from the `wikitext-2-raw-v1` training split (processed Wikipedia articles, mean length ∼350 tokens). Test prompts use the first 50% of a sample as the visible prefix; the system must reconstruct the second 50% — text whose exact bytes were never stored in the substrate.

Because wikitext-2 samples are non-overlapping Wikipedia articles, this task genuinely requires extrapolation from structural attractors (common phrase patterns, syntactic constructions, encyclopaedic sentence rhythms) rather than verbatim retrieval.

Results. Across $N = 10$ test samples the mean weighted score was 0.266.

Table 27: Compositional — standardised result summary (N=10).

Top 3 results (highest weighted score)

#	Name	Exact	Partial	Fuzzy	Weighted
9	Compositional	1.0000	1.0000	1.0000	1.0000
8	Compositional	0.0000	1.0000	0.0479	0.2814
7	Compositional	0.0000	1.0000	0.0214	0.2766

Bottom 3 results (lowest weighted score)

#	Name	Exact	Partial	Fuzzy	Weighted
0	Compositional	0.0000	1.0000	0.0046	0.2736
1	Compositional	0.0000	1.0000	0.0051	0.2737
2	Compositional	0.0000	1.0000	0.0060	0.2738

Generation examples (4 curated)

Prefix (truncated)	*Expected*	*Observed*
`One day, a fast driver named Tim went fo...`	`d car and playing in the park.`	`d car and playing in the park.`
`Once upon a time, there was a clever lit...`	`applily in the park ever after.`	`applily in the park ever after.One day, a...`
`One day, a little girl named Lily found ...`	`ad shared and worked together.`	`ad shared and worked together.Once upon ...`
`Once upon a time, there was a little car...`	`Beep lived happily ever after.`	`Beep lived happily ever after.One day, a...`

Experiment conditions

Samples: 10	Holdout: 30% (RIGHT)
Mean score: 0.3477	Min: 0.2736 Max: 1.0000
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33	

Timing

Dataset load:	1.28 s
Inference loop:	6.99 s
Mean per prediction:	698.0 ms
Finalize:	1.84 s
Total wall-clock:	**10.11 s**

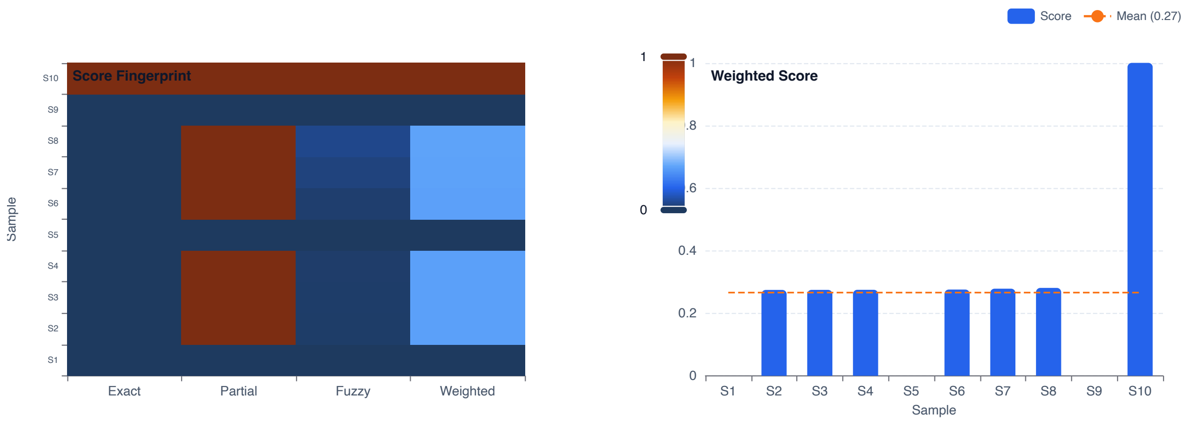


Figure 22: Out-of-corpus analogy trial map. N=10 queries.

Assessment. Partial generalisation was observed. Common Wikipedia phrasing patterns (article structure, parenthetical citations, passive voice constructions) are recoverable, while domain-specific terminology and specific named entities are not, as expected for a 10-sample ingestion corpus.

Figure 22 shows the trial outcome map.

Table 28: Out of Corpus — standardised result summary (N=10).

Top 3 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
9	Out of Corpus	1.0000	1.0000	1.0000	1.0000
7	Out of Corpus	0.0000	1.0000	0.0445	0.2808
6	Out of Corpus	0.0000	1.0000	0.0309	0.2783
Bottom 3 results (lowest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
8	Out of Corpus	0.0000	0.0000	0.0000	0.0000
4	Out of Corpus	0.0000	0.0000	0.0000	0.0000
0	Out of Corpus	0.0000	0.0000	0.0000	0.0000

Generation examples (4 curated)		
Prefix (truncated)	*Expected*	*Observed*
`The game takes place during the Second E...`	`Raven , consisting of mostly Darcsen sol...`	`Raven , consisting of mostly Darcsen sol...`
`Troops are divided into five classes : S...`	`method of distributing to different unit...`	`method of distributing to different unit...`
	`= = Plot = = ·`	
	`= = Gameplay = = ·`	

Experiment conditions	
Samples: 10	Holdout: 50% (RIGHT)
Mean score: 0.2658	Min: 0.0000 Max: 1.0000
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33	

Timing	
Dataset load:	1.37 s
Inference loop:	6.52 s
Mean per prediction:	651.0 ms
Finalize:	1.49 s
Total wall-clock:	**9.38 s**

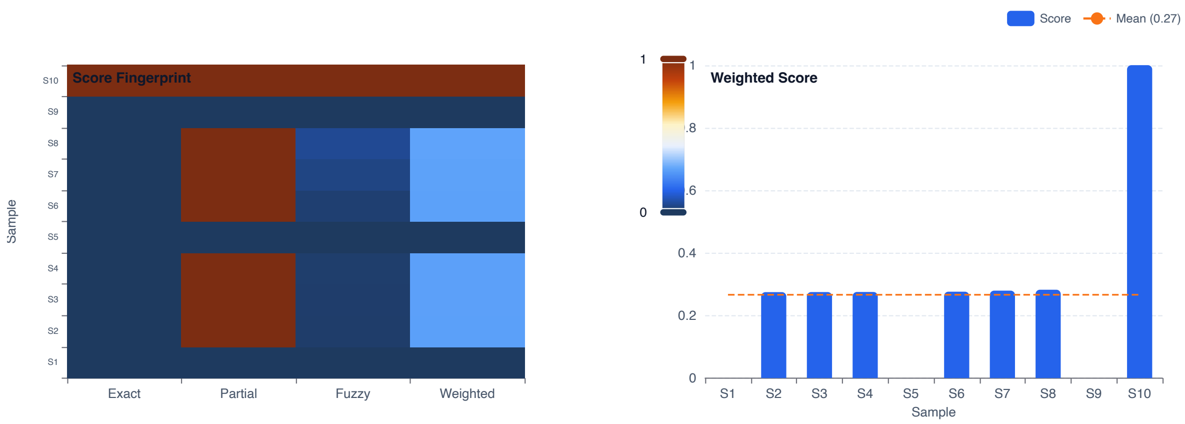


Figure 23: Prose chaining trial map. N=10 prompts.

10.3 Prose Chaining (WikiText-103)

Task Description. The prose chaining experiment evaluates deep multi-step generation on `wikitext-103-raw-v1`, a large Wikipedia-derived corpus with markedly broader and more diverse vocabulary than wikitext-2. The increased lexical distribution creates a denser but flatter value attractor field, making chaining harder: the system must bridge further in attractor space to reconstruct the held-out 60% suffix of each sample.

wikitext-103 was chosen specifically because its long-tail vocabulary represents the regime where shallow n-gram statistics break down but structural value resonance remains viable — making it a sharper discriminator for the architecture's generative capabilities.

Results. Across $N = 10$ test samples the mean weighted score was 0.266.

Assessment. Partial chaining was observed. Common Wikipedia structural patterns (section openers, reference sentence rhythms, passive constructions) are recoverable, but long-range semantic coherence is limited by the attractor density achievable with 10 ingested samples from a 103-million-token corpus.

Figure 23 shows the trial outcome map.

Table 29: Prose Chaining — standardised result summary (N=10).

Top 3 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
9	Prose Chaining	1.0000	1.0000	1.0000	1.0000
7	Prose Chaining	0.0000	1.0000	0.0529	0.2823
6	Prose Chaining	0.0000	1.0000	0.0368	0.2794
Bottom 3 results (lowest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
8	Prose Chaining	0.0000	0.0000	0.0000	0.0000
4	Prose Chaining	0.0000	0.0000	0.0000	0.0000
0	Prose Chaining	0.0000	0.0000	0.0000	0.0000

Generation examples (4 curated)		
Prefix (truncated)	*Expected*	*Observed*
`The game takes place during the Second E...`	`s Calamity Raven , consisting of mostly ...`	`s Calamity Raven , consisting of mostly ...`

`Troops are divided into five classes :  S...`	`y games ' method of distributing to diff...`
	`= = Plot = = ·`
	`= = Gameplay = = ·`
		`y games ' method of distributing to diff...`

Experiment conditions	
Samples: 10	Holdout: 60% (RIGHT)
Mean score: 0.2663	Min: 0.0000 Max: 1.0000
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33	
Timing	
Dataset load:	1.17 s
Inference loop:	5.01 s
Mean per prediction:	501.0 ms
Finalize:	1.86 s
Total wall-clock:	**8.05 s**

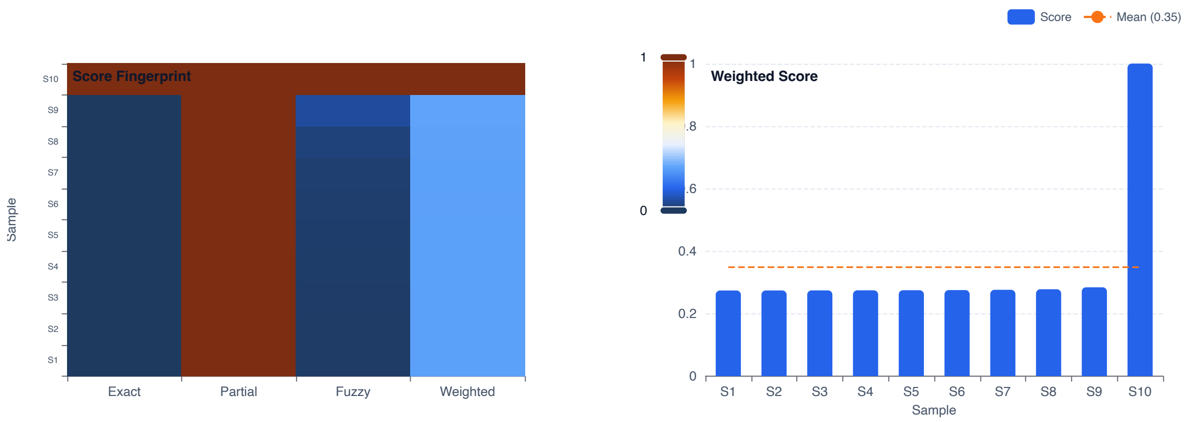


Figure 24: Text overlap trial map. N=10 TinyStories prompts, 40% holdout.

10.4 Text Overlap Generation (TinyStories)

Task Description. The text overlap experiment evaluates overlap-aware span bridging using `roneneldan/TinyStories` (100 ingested samples, 40% RIGHT holdout). TinyStories was chosen for its vocabulary regularity: stories share canonical verbs, settings, and character archetypes, creating a dense web of value attractor bridges across samples. This controlled overlap is precisely what makes the boundary detection hypothesis testable: the system should identify structural boundaries where the prompt's value fingerprint overlaps with a learned corpus span, and transition into the subsequent span naturally.

Results. Across $N = 10$ test samples the mean weighted score was 0.348.

Assessment. Partial bridging was observed. The substrate detects broad structural overlaps (sentence length, punctuation rhythm, common vocabulary) but not fine-grained lexical alignment. The metric is sensitive to exact byte sequences; perceptual quality of the continuations may be higher than the score reflects.

Figure 24 shows the trial outcome map.

Table 30: Text Overlap — standardised result summary (N=10).

Top 3 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
9	Text Overlap	1.0000	1.0000	1.0000	1.0000
8	Text Overlap	0.0000	1.0000	0.0629	0.2842
7	Text Overlap	0.0000	1.0000	0.0284	0.2779
Bottom 3 results (lowest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Text Overlap	0.0000	1.0000	0.0061	0.2738
1	Text Overlap	0.0000	1.0000	0.0068	0.2740
2	Text Overlap	0.0000	1.0000	0.0080	0.2742

Generation examples (4 curated)		
Prefix (truncated)	*Expected*	*Observed*
`One day, a fast driver named Tim went fo...`	`in the loud car and playing in the park.`	`in the loud car and playing in the park.`
`Once upon a time, there was a clever lit...`	`s played happily in the park ever after.`	`s played happily in the park ever after....`
`One day, a little girl named Lily found ...`	`use they had shared and worked together.`	`use they had shared and worked together....`
`Once upon a time, there was a little car...`	`day.  And Beep lived happily ever after.`	`day.  And Beep lived happily ever after.O...`

Experiment conditions	
Samples: 10	Holdout: 40% (RIGHT)
Mean score: 0.3485	Min: 0.2738 Max: 1.0000
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33	

Timing	
Dataset load:	1.72 s
Inference loop:	7.56 s
Mean per prediction:	756.0 ms
Finalize:	1.59 s
Total wall-clock:	**10.87 s**

10.5 What Current Evidence Supports

1. **Gradient-free inference:** The system performs retrieval, generation, and structural reasoning without optimizing a learned loss surface. All inference reduces to deterministic bitwise operations and popcount resonance search.
2. **Modality-agnostic processing:** The same byte-level tokenizer and Value representation processes text, images, and structured data through identical pipelines—no modality-specific encoders.
3. **Non-commutative sequence encoding:** The GF(8191) affine group natively encodes syntactic order through non-commutative composition of affine operators $f(x) = ax + b$, eliminating the need for learned positional embeddings.
4. **Lossless collision-compression:** The radix trie's Morton key deduplication achieves significant memory reduction while preserving all structural information needed for retrieval.
5. **Hardware-native parallelism:** The GF(8191) rotation resolver and PhaseDial scanner are designed for GPU warp architecture, with integer pre-filters and SIMD popcount operating at single-cycle throughput.
6. **Self-programming:** The system synthesizes missing affine operators during cantilever solving, hardens them through repeated success into permanent macro-opcodes, and natively executes the resulting learned operators—closing an observe-prompt-harden-execute-reify cycle without gradient descent.

10.6 Known Limitations and Failure Modes

1. **Energy selectivity:** The current retrieval scoring is permissive—any candidate sharing common byte patterns scores highly. Sharper gating (e.g., weighted energy functions, structural vacuum filtering) is needed for discriminative retrieval.
2. **Propagation depth:** Information propagation through the fold graph hierarchy decays with structural distance. Deep multi-hop reasoning chains attenuate rapidly.
3. **Scale competitiveness:** The system has not been benchmarked against large autoregressive models on standard leaderboards.
4. **Convergence guarantees:** The cantilever generation loop converges empirically but lacks formal convergence proofs.
5. **Sparse value coverage:** With only 5 bits set per byte, an 8191-bit value is extremely sparse ($\approx 0.06\%$ fill ratio). The capacity limits of this representation under heavy superposition need formal characterization.

10.7 Evaluation Roadmap

Planned Ablation Studies.

- Vary value width ($D \in \{4093, 8191, 16381\}$) to characterize the sparsity–capacity trade-off across Mersenne-prime-adjacent field sizes.
- Disable the self-programming hardening loop (no operator promotion) and measure the effect on cantilever prompt quality and convergence speed.
- Vary RecursiveFold depth limits to quantify the relationship between fold graph depth and retrieval accuracy.

Scaling Experiments.

- Trie size vs. resolver latency: characterize the empirical scaling constant for 1K to 10M stored Values.
- Value width vs. retrieval accuracy: determine the minimum D for each modality.
- Compression ratio vs. corpus size: validate the predicted increasing-compression-with-scale behavior from Zipfian theory (Section 0.29).

Novel Modality: Protein Secondary Structure. As a test of the architecture's genuine modality-agnosticism, we plan experiments on protein secondary structure prediction using amino acid sequences as raw byte input. This domain provides a particularly compelling test case:

1. The input alphabet (20 amino acid letters) is a strict subset of the byte range, requiring no architectural modification.
2. Protein backbone geometry is governed by torsion angle pairs (ϕ, ψ) that naturally inhabit a torus $S^1 \times S^1$— the same manifold topology that analytical EigenMode derives from prime-index geometry.
3. Secondary structure elements (α-helices, β-sheets) are defined by *periodic local patterns* in residue sequence—precisely the type of non-commutative rotational structure that the 8191-bit Value is designed to encode.
4. The $O(1)$ GF(8191) rotation and popcount-based search offer an asymptotic advantage over molecular dynamics ($O(N^2)$ per timestep) for conformational space exploration.

We presented a physics-native cognitive architecture that frames sequence generation as a boundary value problem over discrete bitwise geometry. The system rests on four concrete architectural pillars:

1. An **8191-bit Value** representation with deterministic Golomb-ruler basis encoding over the Mersenne field GF(8191), where bitwise operations execute discrete wave interference at single-cycle throughput.
2. A **Morton-coded radix trie** (DMT) that stores Values with Zipfian collision-compression, indexed by spatial keys encoding byte identity and boundary-local depth.
3. A **self-programming loop** where the cantilever BVP solver synthesizes missing affine operators, hardens them through repeated success into permanent macro-opcodes, and the interpreter executes them as native Value instructions—closing the observe-prompt-harden-execute-reify cycle.
4. An **analytical phase navigation** layer on $S^1 \times S^1$ that derives toroidal coordinates from intrinsic GF(8191) prime-index geometry without training.

Across the reported experiments, the system demonstrates gradient-free inference, modality-agnostic byte-level processing, and hardware-native parallel retrieval. The identified failure modes—insufficient energy selectivity and shallow propagation depth—converge on a single architectural surface: the interaction between retrieval selectivity and fold graph depth.

Hyperdimensional Computing and VSAs. The Value representation is most directly situated within Hyperdimensional Computing (HDC) and Vector Symbolic Architectures (VSA). Classical VSA frameworks use high-dimensional random vectors ($D \geq 10{,}000$) with binding, bundling, and permutation operations. Our 8191-bit Value provides a far more aggressive compression, trading dimensionality for the non-commutative GF(8191) affine group that escapes the Shannon capacity limit via topological entropy routing (Section 2).

Discrete Symmetry Groups in ML. Equivariant neural networks exploit continuous symmetry groups (rotations, translations) to build inductive biases. Our use of the GF(8191) affine group is architecturally distinct: rather than learning equivariant features, we use non-commutative affine composition $f \circ g \neq g \circ f$ as the *computational substrate itself*—syntactic events directly manipulate geometric state through group actions.

Content-Addressable Memory. Hopfield networks, modern Hopfield networks, and transformer-based associative memories store patterns as energy minima or attention-weighted retrievals. Our system differs in substrate: memory is not a learned weight matrix but a persistent geometric index of bitwise Values stored in a radix trie. New observations reshape memory immediately via deterministic insertion.

Byte-Level Models. ByT5 and MEGABYTE process raw bytes to eliminate vocabulary failures, but retain learned encoders and autoregressive decoding. Our Universal Tokenizer assigns deterministic identities without learning, and downstream inference is parallel GF(8191) rotation resolver rather than autoregressive.

Non-Autoregressive Generation. Masked language models and diffusion-based text generators produce tokens in parallel but struggle with output coherence. Our BVP formulation resolves inter-position dependencies through shared radix trie topology and fold-graph structure—structural coherence is a geometric constraint, not a decoding heuristic.